

Ember: the long life of a $0.1M_{\odot}$ star

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Summary

Ember is a one-dimensional stellar-evolution code written to follow a $0.1M_{\odot}$ star from contraction, through hydrogen burning lasting several trillion years, to a cooling helium white dwarf and its conditional fate if nucleons decay.

A continuous calculation starts on the Hayashi track and uses one program from deuterium burning onward, with composition-dependent atmospheres and hot microscopic transport. It reaches 3.562Tyr at 3259 K. The centre is radiative, the outer envelope remains convective, and surface hydrogen is $X = 0.1757$. The outer boundary remains at optical depth 100. A fixed-metal atmosphere comparison reaches 3.078Tyr; neither continuous calculation has yet formed the late burning shell.

A comparison calculation, started from a static hydrogen-burning model on its own clock, includes gravitational settling of helium and metals and reaches 4.002Tyr. Whole-star convection ends near 3.55Tyr, central hydrogen falls below one part in a thousand near 3.65Tyr, and luminosity and surface temperature pass shallow maxima ($0.006513L_{\odot}$; 4754 K) before the model turns down at 4612 K, still 97% powered by hydrogen burning. A companion calculation without settling is about twice as luminous as the 1997 model of Laughlin, Bodenheimer and Adams at the same mass and reaches each burning stage in about 60% of the time; MESA runs with grey atmospheres show the same sequence of events.

When the outer boundary of the settling model is switched to a published pure-hydrogen white-dwarf atmosphere, a helium-3 shell flash develops. It appears on every mass grid and mixing treatment tried, but every computed history descends from one post-switch state, the shell was fading before the switch, and no other calculation shows it. Its peak, its duration and whether it occurs with the original boundary are not established; the continuations through it are provisional, and the control with the original atmosphere is pending.

The continuous settling calculation still needs atmosphere coverage through cooling below 4600 K, coupled grain formation, hydrogen-free atmospheres and a cold helium equation of state. Grey-atmosphere comparisons already reach lower temperatures. The paper ends with a conditional timeline to the loss of the remnant.

1 The question and the calculations

A star of $0.1M_{\odot}$ burns hydrogen for several trillion years. Laughlin, Bodenheimer and Adams [9] (LBA97) followed its evolution into white-dwarf cooling with a grey atmosphere and the opacities and reaction rates of the 1990s. The aim of Ember is to follow such a star from formation through hydrogen exhaustion, the transition to a helium white dwarf, cooling through 1000, 500 and 100 K, and the conditional stages that follow: encounters, accretion, dark-matter heating, and disappearance if nucleons decay. The remit is to use the simplest model that resolves each question and to keep numerical uncertainty below about a tenth of the physical variation that matters.

The paper reports five calculations. Their clocks differ and they have not been joined into one track.

- The **continuous calculation** starts from a Hayashi-track model and runs deuterium, proton–proton and CN burning, plasma neutrino losses and convective mixing in one program. With hot microscopic

transport and composition-dependent atmospheres it reaches 3.562Tyr. Age is measured from the contracting starting model (Section 3).

- **Track S** (settling) starts from a static hydrogen-burning model at age zero and follows hydrogen, helium-3, carbon, nitrogen and a moving metal group with microscopic diffusion and settling. It reaches 4.002Tyr and is the reference long-term calculation.
- **Track F** (fixed metals) uses the same network and diffusion with the metal abundance held fixed; it reaches 3.704Tyr.
- **Track P** (proton–proton only) omits carbon–nitrogen burning and microscopic diffusion; it reaches 3.890Tyr and is the basis of the LBA97 comparison.
- The **flash continuations** F1 and F2 start from Track S after its outer boundary is replaced by a pure-hydrogen white-dwarf table. They are provisional (Section 5).

Sections 2–4 give the model, the contraction results and the hydrogen-burning results. Section 5 separates what is established about the helium-3 flash from what is not. Section 6 compares with LBA97, with MESA and with observed stars. Section 7 lists the missing physics that can change the interpretation, Section 8 gives the conditional lifetime timeline, and Section 9 records cost and reproducibility. The appendix holds reusable implementation detail. Every number in the text is a calculated value unless labelled as a literature value or a conditional estimate. Table 1 summarizes the status of each calculation.

Table 1: Status of the calculations reported here.

Calculation	Clock	Last computed state	Status
Continuous calculation	from Hayashi model	3.562Tyr, 3259 K, surface $X = 0.1757$	radiative core, convective envelope
Fixed-metal atmosphere comparison	from Hayashi model	3.078Tyr, 3130 K	complete comparison; fully convective
Track S	static H-burning model	4.002Tyr, 4612 K	complete to atmosphere limit
Track F	static H-burning model	3.704Tyr, 3702 K	complete (comparison)
Track P	static H-burning model	3.890Tyr, 5546 K	complete (comparison)
Flash continuations F1, F2	from post-switch model	5.744 Myr; 42.48 Myr	provisional; control pending
Continuous settling calculation below 4600 K	—	not computed	requires new atmospheres and cold EOS

2 The model

All calculations hold the baryonic mass fixed at $0.1M_{\odot}$ and omit rotation, magnetism, mass loss, accretion and external heating. The initial hydrogen mass fraction is $X = 0.7$ and the metal fraction $Z = 0.02$, with the relative metal abundances of the Grevesse–Sauval (GS98) mixture and the remainder in ^4He . Tracks S, F and P start with no ^3He ; the continuous calculation starts with initial deuterium. Table 2 distinguishes the physical assumptions of the comparison tracks from those selected for the continuous calculation.

Table 2: Physical ingredients by calculation.

Component	Tracks S, F, P and flash continuations	Continuous calculation
Starting model	Static, fully convective hydrogen-burning model at age zero: $R = 0.1259R_{\odot}$, $T_{\text{eff}} = 2766$ K, $L = 0.0008362L_{\odot}$, central density 381.7 g cm^{-3} .	Fully convective Hayashi-track model: $T_{\text{eff}} = 2864$ K, $R = 1.594R_{\odot}$, central temperature 0.5398 MK, with initial deuterium.
Nuclear reactions	Reduced ppI/ppII network with ${}^3\text{He}$ out of equilibrium, Solar Fusion II rates [24] and finite-degeneracy Salpeter–Van Horn screening. Tracks F and S add a time-dependent CN cycle (${}^{12}\text{C}$, ${}^{13}\text{C}$, ${}^{14}\text{N}$) with the ${}^{14}\text{N}$ capture rate of Solar Fusion III [25]; in Track S nuclear captures change the metal mass. Track P has no CN burning. Pep, hep, ppIII and the oxygen branches are absent in all.	Initial deuterium, pp and closed CN burning with Solar Fusion III rates for both networks and the same screening prescription. Pep, hep, ppIII and oxygen branches absent.
Heat transport	Mixing-length convection with the Ledoux criterion; radiative opacity plus electron conduction from Ioffe/Cassisi mixtures; plasma-neutrino losses [13]. Other thermal-neutrino processes are omitted.	Ledoux mixing-length convection, radiation, plasma-neutrino losses and conductive heat transport. In hot layers the screened kinetic calculation supplies the conductive coefficient; it joins the cool-envelope Ioffe/Cassisi coefficient smoothly between 2 and 3 MK.
Composition transport	Track P: instantaneous mixing within each convective region, slow mixing in marginally stable layers, no microscopic diffusion. Track F: microscopic diffusion of H, ${}^3\text{He}$, C and N with the metal inventory fixed; chemical forces from the equation-of-state free energy; electron collisions averaged over the finite-temperature Fermi distribution. Track S: as F, with all heavy elements moving as one group with distinct charges, and the equation of state and opacity depending on the total metal abundance. Flash continuations: finite mixing $D = v\ell/3$ in hot convective regions; semiconvective and thermohaline coefficients set to zero.	Instantaneous whole-star mixing while the star is fully convective, with the heat carried by changing composition included after deuterium exhaustion. Screened H/He/metal species fluxes act across radiative boundaries at temperatures of at least 2 MK; the metal law assumes full ionization. Cooler radiative species transport is unsupported. The heat law, rather than the species flux, uses the smooth 2–3 MK join.
Equation of state	FreeEOS 3.0 free-energy tables with numerical electron integrals, interpolated through a shared Helmholtz potential so that pressure, energy and composition responses stay consistent. Track P: 72 composition tables. Track F: 152 tables at 38 hydrogen and four ${}^3\text{He}$ fractions. Track S: a variable-metal family from $Z = 0$ to 0.08, extended to 0.16 for the metal-enriched core. None describes the cold, dense remnant.	One interpolated potential with planes at $Z = 0, 0.005, 0.02, 0.04$ and 0.16, twelve hydrogen-share coordinates and four ${}^3\text{He}$ -share coordinates; all 512 zones support the required composition derivatives.
Opacity	AESOPUS 2.1 gas opacity at low temperature (no grains) and Los Alamos ATOMIC/TOPS opacity at high temperature [4, 5], including hydrogen-free mixtures above 5.012×10^5 K. Composition continuations in X and Z are given in Appendix A.	Same tables and continuations.

Component	Tracks S, F, P and flash continuations	Continuous calculation
Atmosphere	TLUSTY/SYNSPEC plane-parallel LTE atmospheres with frequency-resolved absorption, scattering and collision-induced absorption, no condensates, joined to the interior at Rosseland optical depth 100. Track P: 220 atmospheres over $X = 0.1$ – 0.7 , ${}^3\text{He}$ 0 – 0.12 , through 5600 K. Track S: additional families down to $Z = 10^{-8}$ and $1 - X = 10^{-5}$, 4600 – 6000 K, $\log g$ to 5.95 , with a local continuation to 6.1 . Flash continuations: the pure-hydrogen white-dwarf table distributed with MESA [3, 2], joined at optical depth 25.12 and extrapolated below $\log g = 5.5$.	A composition-dependent family at $X \geq 0.65$, ${}^3\text{He} \leq 0.0075$, fixed $Z = 0.02$, joined at optical depth 100. The continuous metal-dependent selection joins measured families through nearly pure hydrogen, including trace helium, at the same optical depth. Cool, high-gravity rows include inferred continuation; the selected source range ends at 4600 K and $\log g = 6.1$.
Mesh and time integration	512 mass points, fixed for Tracks S, F and P; flash grids refined locally to 578–2115 points. Structure, burning and mixing are solved together implicitly. Every accepted step is compared with two half steps; species inventories, inert-metal mass, nuclear rest-mass release and the first law are checked independently. Saved models restart exactly.	512 zones; the same step-doubling check; independent isotope, mass-defect and first-law audits on every interval; exact restart replay. Radiative-core evolution and restart are registered tests.

For Tracks S, F and P the relative time-error tolerances are 0.0003 for radius, density and temperature, 0.003 for local ${}^3\text{He}$ and 0.001 for the other species; the abundance solver converges to 10^{-10} . Tightening the structure and ${}^3\text{He}$ tolerances threefold over a matched 5 Gyr interval changes luminosity by 0.002872% and surface temperature by 0.01705 K, while loosening the structure tolerance to 0.001 over 2 Gyr changes luminosity by 1.078% and temperature by 3.114 K; the tracks therefore keep the original settings. The continuous calculation uses a structure tolerance of 10^{-4} and an energy tolerance of 0.005. Local composition solves converge to 10^{-15} in a single instantaneously mixed region and 10^{-12} in a stratified star; the separate global abundance-balance tolerance remains 10^{-14} .

Throughout the paper a *paired comparison* means two calculations started from the same model, differing in one setting, and compared at the same age. Such comparisons measure sensitivity to that setting over the stated interval; they are not error bounds for a complete lifetime, and the text says so wherever the distinction matters.

Replacing the Solar Fusion II rates by Solar Fusion III with everything else fixed raises the initial luminosity by 0.7454% and the radius by 0.2164%. FreeEOS omits potassium (mass fraction 4.56×10^{-6}). The interior equation of state, atmosphere chemistry, opacity and conduction come from different sources; their mutual consistency is checked at sampled states (Appendix A) rather than guaranteed.

3 Contraction and the continuous calculation

The contraction calculation starts from a fully convective model at $T_{\text{eff}} = 2864$ K and $R = 1.594R_{\odot}$. Initial deuterium burns first and is exhausted; proton–proton burning then rises to supply the surface luminosity (Figure 1). Near 1 Gyr the model has $T_{\text{eff}} = 2763$ K, $R = 0.1265R_{\odot}$, $L = 0.0008401L_{\odot}$, central temperature 4.622 MK and $\log g = 5.234$, with nuclear burning supplying 99.09% of the light. The abundances are uniform, $X = 0.6998$ and $X_3 = 0.0002376$. The first 1000 accepted intervals reach this state without rejections; each interval passes the time-error, isotope, mass-defect and first-law checks.

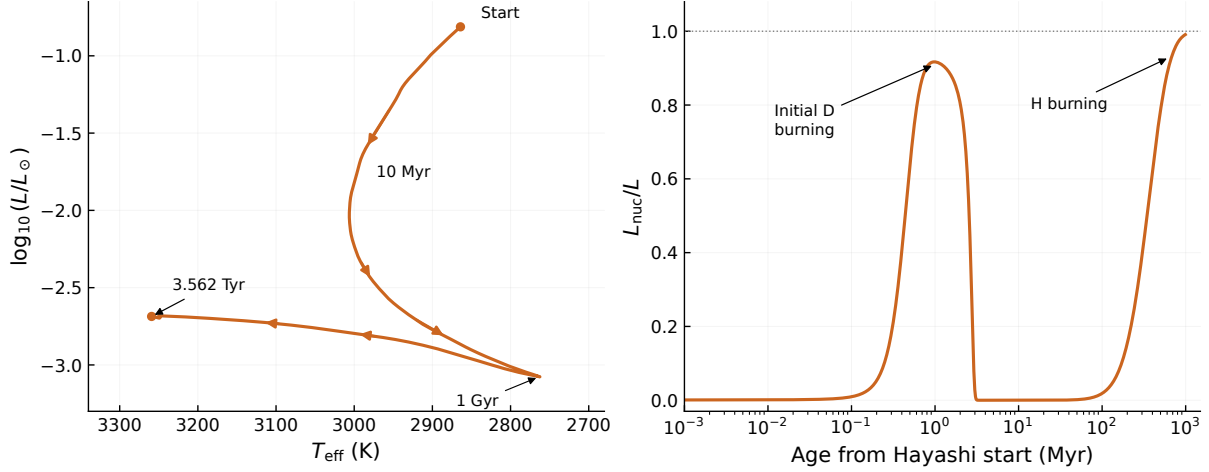


Figure 1: Continuous evolution of the $0.1M_{\odot}$ model from the Hayashi track. Left: the HR trajectory through 3.562Tyr; arrows show increasing age. Right: nuclear power divided by surface luminosity during the first Gyr; the early maximum is deuterium burning. Age is measured from the contracting starting model. These panels use the same continuous calculation, whose outer boundary remains at optical depth 100. Tracks S, F and P have their own initial models and clocks.

During deuterium burning the convective travel time is 0.9535–1.415 yr, compared with local burning times of about 2000 yr or longer. The omitted heat carried by composition redistribution remains below 0.01% of the local luminosity. After deuterium exhaustion that heat is included. Instantaneous mixing is checked against the abundance gradients required to carry the reconstructed material flux within each convective region. Microscopic species exchange is followed explicitly across hot radiative boundaries when they form; cooler radiative exchange needs additional transport physics.

Comparison with observed stars. Table 3 compares the continuous model near 1 Gyr with three measured objects [6, 7, 8]. The closest comparison is the eclipsing companion EBLM J2114–39 B, whose mass agrees with $0.1M_{\odot}$ and whose radius is about 1.2% smaller than Ember’s, within its uncertainty. Its measured metallicity is $[\text{Fe}/\text{H}] = -0.001 \pm 0.032$; a precise comparison would also match the initial helium and metal abundances. TRAPPIST-1 and Proxima bracket Ember in radius, temperature and luminosity, but their masses rest on a mass–luminosity relation and on stellar models respectively. Ten billion years of hydrogen burning changes Ember’s luminosity by less than 0.4% and its temperature by less than 2 K, so main-sequence age does not affect the comparison. The agreement supports the starting structure without establishing percent-level accuracy or selecting among atmosphere and interior choices.

Table 3: Continuous model near 1 Gyr and observed low-mass stars.

Star	Mass ($M_{\odot}$)	Radius ($R_{\odot}$)	T_{eff} (K)	Luminosity ($L_{\odot}$)
Ember, near 1 Gyr	0.1	0.1265	2763	0.0008401
EBLM J2114–39 B	0.0993 ± 0.0033	0.1250 ± 0.0016	—	—
TRAPPIST-1	0.0898 ± 0.0023	0.1192 ± 0.0013	2566 ± 26	0.000553 ± 0.000019
Proxima	0.120 ± 0.003	0.146 ± 0.007	2980 ± 80	0.00151 ± 0.00008

Continuous hydrogen burning. With composition-dependent atmospheres and hot microscopic transport, the Hayashi-started calculation reaches 3.562Tyr in 7678 accepted intervals and 453 rejected

trials. At that point surface $X = 0.1757$, $T_{\text{eff}} = 3259$ K, $R = 0.1423R_{\odot}$, $L = 0.002059L_{\odot}$, and the central temperature is 8.593 MK. Nuclear burning supplies 99.9% of the surface luminosity. A radiative shell first appears at 3.558Tyr, separating the convective centre from the envelope. The centre subsequently becomes radiative; the radiative interior now contains 35.64% of the mass, beneath a convective envelope. Microscopic diffusion is active across the radiative interfaces, while convection mixes each connected convective region.

A fixed-metal atmosphere comparison reaches 3.078Tyr, 3130 K and $X = 0.2629$, also without rejected intervals. The composition-dependent selection spans the saved settling structures, including their trace-helium and nearly pure-hydrogen surfaces. All sources join at optical depth 100; the coolest, highest-gravity rows include inferred values, and support below 4600 K remains to be completed. The continuous calculation has not yet tested whether a helium-3 flash occurs. Tracks S, F and P supply the late hydrogen-burning comparisons in Section 4.

A 50 Gyr time-resolution comparison in the fully convective phase supports an absolute species time-error tolerance of 10^{-7} : it uses 25 intervals instead of 73 at 10^{-8} , with maximum relative density difference 7.451×10^{-6} and helium-3 inventory difference 1.502×10^{-5} . In the radiative-core transition, a matched 100 Myr comparison supports 10^{-6} : the largest temperature difference is 3.991×10^{-6} relative and the total helium-3 difference is 2.978×10^{-6} relative. The structure tolerance remains 10^{-4} , and all conservation checks are unchanged. These comparisons do not establish convergence during a flash.

4 Hydrogen burning through four trillion years

4.1 The three long-term tracks

Tracks S, F and P start from the same kind of static hydrogen-burning model and differ in nuclear network and composition transport (Table 2). Figure 2 shows all three in the HR diagram with the LBA97 track; Figure 3 shows the composition and convection histories. Table 4 lists the computed milestones by track.

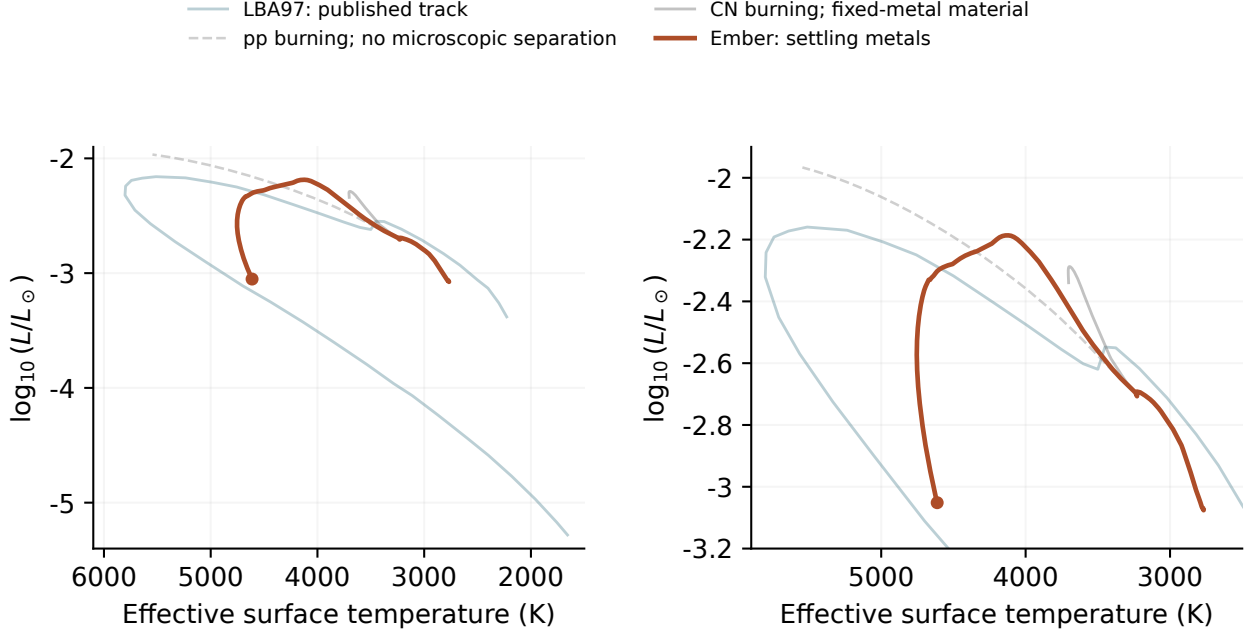


Figure 2: HR diagrams of Track S (orange, through 4.002Tyr), Track F (solid gray, through 3.704Tyr), Track P (dashed gray, through 3.890Tyr) and the published LBA97 track (pale blue, read from its Figure 1 without age rescaling). Left: full range; right: the hydrogen-burning portion. Track P differs from S in both nuclear and transport physics, so the separation between them is not the effect of settling alone. The small luminosity dip near 3230 K accompanies the formation of an interior radiative layer.

Table 4: Computed milestones. Ages in Tyr on the long-term clock; “—” means not reached or not recorded for that track.

Event	Track S	Track F	Track P
Maximum central ^3He (mass fraction)	—	—	0.7103 (0.1003)
Loss of full convection (central X)	$\simeq 3.55$	3.549 (0.1744)	3.546 (0.1751)
Centre becomes radiative	3.551	3.551	—
Central $X = 10^{-3}$	3.650	3.661	—
Central $X = 10^{-4}, 10^{-5}$	3.660, 3.767	—	—
Luminosity maximum ($L_{\odot}$; T_{eff})	3.673 (0.006513; 4128 K)	3.677 (0.005163; 3695 K)	—
Surface-temperature maximum	3.838 (4754 K)	3.689 (3705 K)	—
Last computed model	4.002	3.704	3.890
T_{eff} (K)	4612	3702	5546
central X	3.528×10^{-8}	1.309×10^{-4}	8.157×10^{-4}
surface X	0.9982	0.8905	0.1730
convective mass fraction	0.01553	—	0.02565
nuclear share of luminosity	97.28%	—	—

At its endpoint Track S has $R = 0.04667R_{\odot}$, $L = 0.0008881L_{\odot}$, central temperature 6.443 MK, central density $3.982 \times 10^4 \text{ g cm}^{-3}$, central metal fraction 0.1312, surface metal fraction below 10^{-40} , and $0.003797M_{\odot}$ of hydrogen remaining. Surface temperature has declined by 141.5 K over the 164.3 Gyr since its maximum, but hydrogen burning still supplies 97.28% of the light; the model has turned toward cooling but is not yet on a passive cooling sequence.

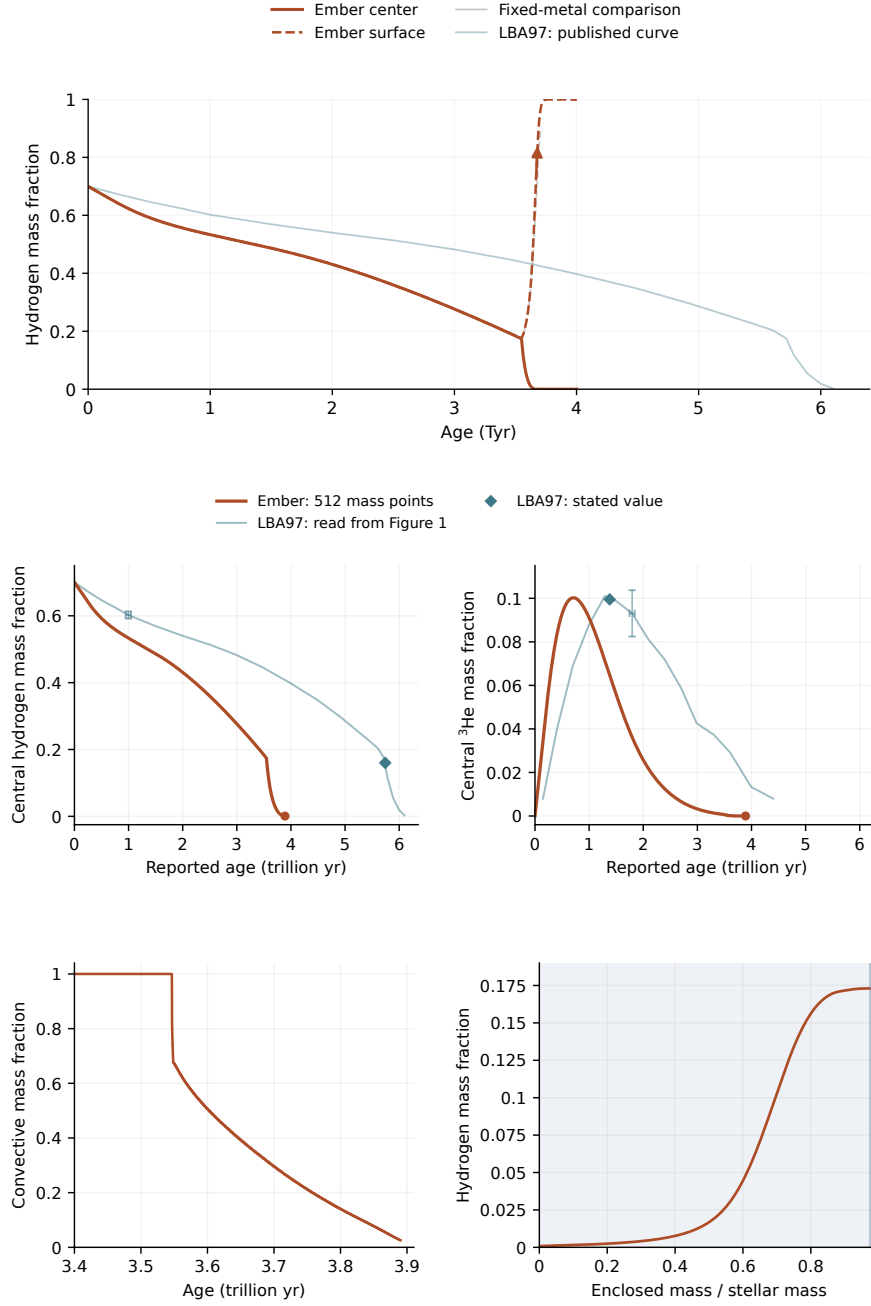


Figure 3: Composition and convection histories. Upper: central (solid) and surface (dashed) hydrogen for Track S (orange) and Track F (gray), with the LBA97 curve in pale blue; the arrow follows increasing age along the Track S surface curve. Lower: hydrogen and ^3He in Track P (orange) against the LBA97 inset (pale curves, read from the printed figure; diamonds mark LBA97's stated ^3He maximum and the hydrogen abundance at radiative-core formation). Crosses show the reading precision of four image pixels, about 0.05Tyr and 0.01 in mass fraction. Ages are shown as reported, without rescaling. Lower: Track P's convective mass fraction, declining from unity near 3.546Tyr to 0.02565 at 3.890Tyr, and its hydrogen profile at 3.890Tyr; the shaded core carries heat by radiation and conduction.

4.2 The end of full convection and the conductive core

Track P first departs from full convection near 3.546Tyr at central $X = 0.1751$: a radiative layer separates the convective centre from the envelope. By 3.560Tyr the convective mass fraction has fallen to 0.6277 and central hydrogen to 0.1518 while the surface remains at 0.1747, so the star is no longer mixed as a whole. In Track F the same transition occurs near 3.549Tyr (3230 K, central $X = 0.1744$) and the central convective region disappears near 3.551Tyr. At that transition carbon-12 depletion is 0.02652, with 0.02524 accumulated as carbon-13 and only 0.001275 converted to nitrogen: the two proton captures cannot be treated as an instantaneous conversion of carbon to nitrogen during this phase.

At 3.890Tyr Track P has a core containing 97.43% of the mass that carries heat by radiation and conduction beneath a convective envelope. Central hydrogen is 8.157×10^{-4} while the envelope remains near 0.1730. Conduction supplies 91.29% of the central heat flux; in Track S at 3.688Tyr electron conduction supplies 99.79%, with conductive and radiative coefficients of 1.058×10^{14} and 2.221×10^{11} erg cm⁻¹ s⁻¹ K⁻¹ and a central degeneracy parameter of 15.39. The composition gradient also suppresses convection. “Fully convective” describes where convection occurs, not the fraction of the luminosity it carries; radiation, conduction and convection share the heat flow continuously, and the mixing-length treatment makes the convective share go to zero at a stability boundary. Chemical mixing uses the stronger approximation that each connected convective region becomes uniform quickly.

During the transition in Track F, a mainly radiative layer at enclosed mass fractions 0.1770–0.2391 separates a convective core from the envelope. Conduction supplies 4.4–4.8% of its diffusive heat flow, yet removing conductivity on that fixed structure would make the layer unstable under the Ledoux criterion: conduction matters for the convection threshold even where radiation carries most of the heat.

4.3 What settling and diffusion change

At their respective luminosity maxima, Track S is 26.13% more luminous and 10.03% smaller than Track F (Table 5), with nearly equal central temperatures. Figure 4 compares the structures. Near 3.673Tyr the layer enclosing 80% of the mass is hotter and denser with settling, 8.260 rather than 7.402 MK and 489.6 rather than 245.0 g cm⁻³, and poorer in hydrogen, $X = 0.1726$ rather than 0.2712; integrating the reaction rates gives 27.25% more nuclear power. Track S has consumed 0.4987% more hydrogen than Track F at this point (80.83% against 80.43% of the initial inventory). CN reactions contribute 1.286% of its nuclear power. The coupled tracks do not isolate the separate contributions of atmospheric opacity, interior material properties and diffusion.

Table 5: Structure at the luminosity maximum.

	Track F	Track S
Age (Tyr)	3.677	3.673
Luminosity ($L_{\odot}$)	0.005163	0.006513
Effective temperature (K)	3695	4128
Radius ($R_{\odot}$)	0.1753	0.1578
Central temperature (MK)	11.90	11.97
Remaining hydrogen ($M_{\odot}$)	$\simeq 0.01370$	0.01342
Convective mass fraction	0.05536	0.03339

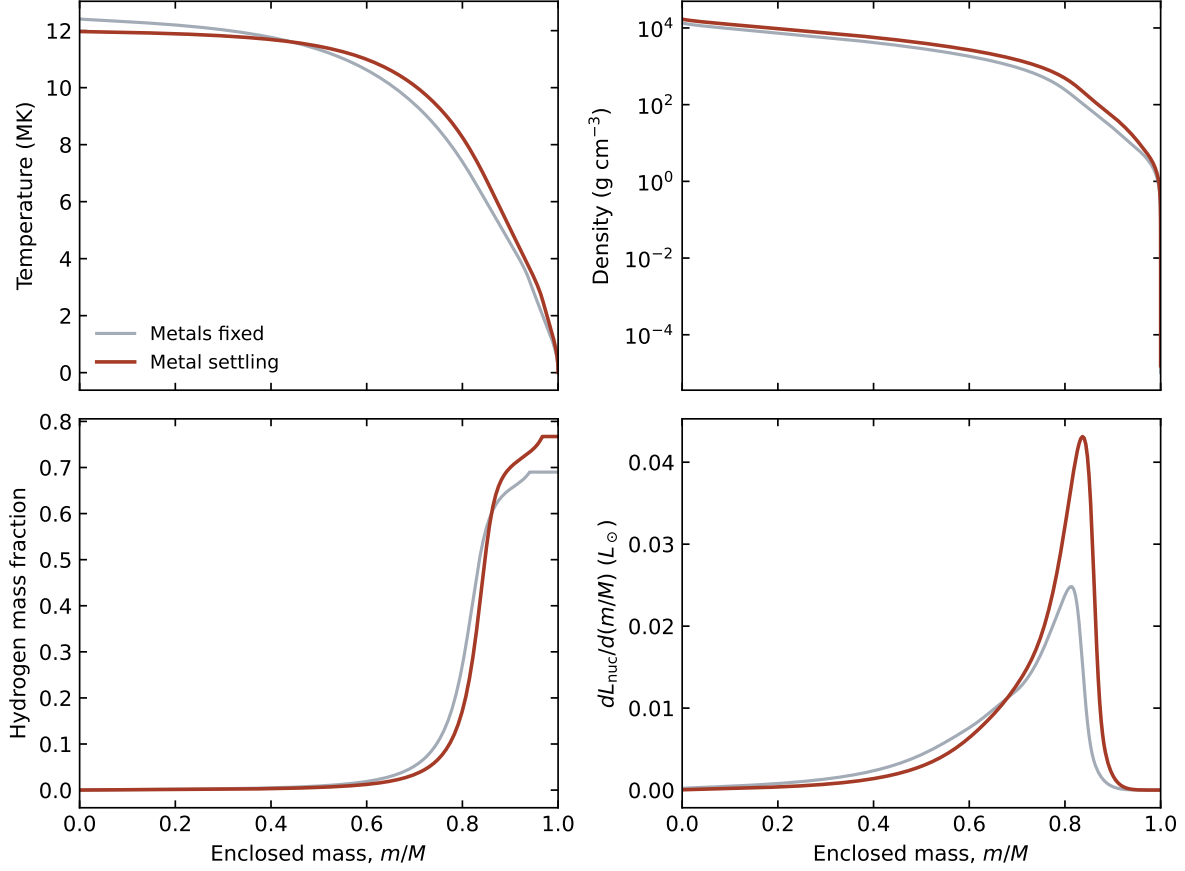


Figure 4: Temperature, density, hydrogen abundance and nuclear power per unit fractional mass at the Track S luminosity maximum, compared with a Track F structure of nearly the same age (0.1837 Gyr later; not the Track F maximum). The strongest burning occurs outside the depleted centre. The curves compare actual structures; exchanging their profiles would not produce equilibrium models.

Diffusion alone. A paired comparison from the same 3.560Tyr Track F structure, evolved 50 Gyr with and without microscopic drift, gives central $X = 0.02300$ with drift and 0.06864 without, surface $X = 0.2907$ and 0.1847 , and effective temperatures of 3404 and 3302 K. Total remaining hydrogen differs by only 0.2767% , but the diffusive calculation burns 4.241% more hydrogen during the interval and ends 20.10% more luminous. Over this interval diffusion mainly redistributes fuel and changes the structure that burns it. Ember’s drift speeds at 14 weakly degenerate interfaces are 0.8163 – 0.9258 of the classical Thoul–Bahcall–Loeb values [18]; the degenerate core needs the actual gravity and electron thermodynamics [23].

For a nearly homogeneous radiative centre, a drift speed of order $Dm_p g/(kT)$ gives a separation time

$$t_{\text{sep}} \sim \frac{kT}{4\pi G \rho m_p D}, \quad (1)$$

of order 100 Gyr at $T \sim 9$ MK, $\rho \sim 300$ g cm $^{-3}$ and $D \sim 0.7$ cm 2 s $^{-1}$. Separation therefore competes with the final core-burning phase while having little effect during the much longer fully convective phase. The residual-fuel scale

$$t_{\text{fuel}} \sim 10^{12} \text{ yr} \left(\frac{\Delta M_{\text{H}}}{10^{-3} M_{\odot}} \right) \left(\frac{L}{10^{-4} L_{\odot}} \right)^{-1} \quad (2)$$

shows why a central hydrogen threshold alone does not fix the duration of shell burning: the energy is available only if hydrogen reaches layers hot enough to burn, and diffusion controls that supply. LBA97 held the envelope composition nearly fixed after the radiative core formed and included no microscopic diffusion; its final whole-star hydrogen fraction is slightly above 0.01 while its envelope retains $X = 0.155$. The tracks here differ from it in exactly the process that decides how much of that envelope hydrogen ever burns.

Settling of metals. In Track S the surface-connected convective envelope shrinks from 4.716% of the mass at 3.664Tyr to 2.013% at 3.727Tyr while its hydrogen fraction rises from 0.6804 to 0.9945; helium and metals diffuse downward through its base and the envelope ends nearly metal-free. A passive calculation following C, N, O, Na, Mg, Si, Ca and Fe on the Track F structures from 3.551 to 3.704Tyr, with the published collision coefficients [18], leaves surface abundances at 0.06213–0.08217 of their initial values with classical heat flow and 0.2491–0.3076 with heat flow suppressed; the two treatments are alternative approximations, not an uncertainty band. Halving the time interval changes the surface results by at most 0.2078%. Metals accumulate in the conductive core, reaching $Z = 0.1312$ at the Track S endpoint. Accretion could maintain surface metals against settling [20]; the isolated model has none.

Removing metals from the atmosphere changes the boundary condition appreciably. At $X = 0.85$, 3800 K and $\log g = 4.9$, reducing carbon and nitrogen to 0.1531 and 0.1541 of their initial abundances (the Track F surface values at 3.697Tyr) changes the joining temperature by +3.314% and the gas pressure by -4.205% ; halving all metals changes them by -8.606% and $+53.69\%$. Applied as fixed boundary factors to the same 3.694Tyr structure for 100 Myr, the two changes give $\Delta T_{\text{eff}} = -30.38$ K and $+142.9$ K, $\Delta L/L = -1.019\%$ and $+5.041\%$, and $\Delta R/R = +1.142\%$ and -4.983% . These establish an appreciable surface-temperature response; the actual response in Track S comes from the transported metal abundance and the metal-dependent atmosphere families.

4.4 Measured sensitivities

Table 6 collects the largest effect found for each class of numerical or table approximation used along Track S. Each entry is a paired comparison over the stated interval; none is a bound on the full lifetime. The one entry that is not small is the absolute normalization of the pure-hydrogen boundary, which is the subject of Section 5.

Table 6: Largest measured sensitivities along Track S.

Approximation tested	Largest table-level discrepancy	Largest stellar response (interval)
Atmosphere interpolation between solved columns (all families, Appendix Table 12)	2.031% in T , 4.656% in P at the joining depth	12.74 K, 0.3110% in L (30 Gyr)
Gravity continuation of the boundary, $\log g = 5.95\text{--}6.1$	0.1878% in T , 2.391% in P against nine solved atmospheres	8.697 K, 0.5101% in L for $\pm 0.5\%/ \pm 5\%$ offsets (50 Myr); 60.96 Gyr of track lies inside
Temperature continuation of the boundary to 4600 K	solved 4600 K atmosphere agrees within 0.05%	17.54 K, 1.026% in L for $\pm 1\%/ \pm 10\%$ offsets (50 Myr)
Pure-hydrogen boundary: TLUSTY against the MESA table at 4600 K, $\log g = 5.9$	2.594% in T , 48.69% in P	about 241 K in 1 Myr (not converged; Section 5)
Molecular opacity continuation below $Z = 0.01$	up to 9.237% at $Z = 0.003$ against withheld sources	0.002433% in L , 0.009728 K between linear and logarithmic forms (10 Gyr)
Radiative opacity $\pm 10\%$ in the depleted envelope	—	0.6315% in L , 3.216 K (1 Myr)
Warm interior opacity $\pm 15\%$ (convective layers)	withheld-source differences up to 11.01%; OPLIB 31.52–35.21%	below 10^{-10} in L and R (1.25 Gyr)
Dense-core radiative opacity halved or doubled	—	no significant change (25 Myr); conduction carries the heat
Variable-metal equation of state	0.7642% in c_P , 1.037% in Γ_1 against FreeEOS	below 0.001 K (10 Myr)
Core equation of state to $Z = 0.16$	0.07726% in material state	0.0004595 K, 0.0002080% in L (25 Myr)
Time-error tolerances threefold tighter	—	0.002872% in L , 0.01705 K (5 Gyr)
Structure tolerance loosened to 0.001	—	1.078% in L , 3.114 K (2 Gyr); rejected

5 The helium-3 shell flash: a provisional result

5.1 Boundary condition and ignition

Track S ends at 4.002Tyr because its TLUSTY atmosphere families stop at $\log g = 6.1$. To continue, the outer boundary was replaced at that point by the pure-hydrogen white-dwarf table distributed with MESA [3, 2], joined at optical depth 25.12, with the interior composition and all evolution checks retained. The two prescriptions are not equivalent: at 4600 K and $\log g = 5.9$ the table gives a joining temperature 2.594% higher and a gas pressure 48.69% lower than an independently converged TLUSTY atmosphere, and neither trace helium, the joining depth (sampled between optical depth 168 and 4775) nor the atmosphere convection prescription (ML2 with $\alpha = 0.6$ and 1.0 change the joining values by at most 0.04571% and 0.2736%) removes the difference. The continuation is therefore a change of boundary condition, not a refinement.

Figure 5 shows the response. The surface temperature first falls to 4362 K, the radius grows by 36.02%, and after 0.7546 Gyr, at 4.003Tyr and 4632 K, the ratio of nuclear to photon luminosity has risen to 2366: a helium-3 burning shell near $m/M = 0.96$ has become convective. The reaction ${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$ consumes the accumulated helium-3 and returns hydrogen. Before the switch the shell was fading, and

every flash history computed so far descends from this one post-switch state.

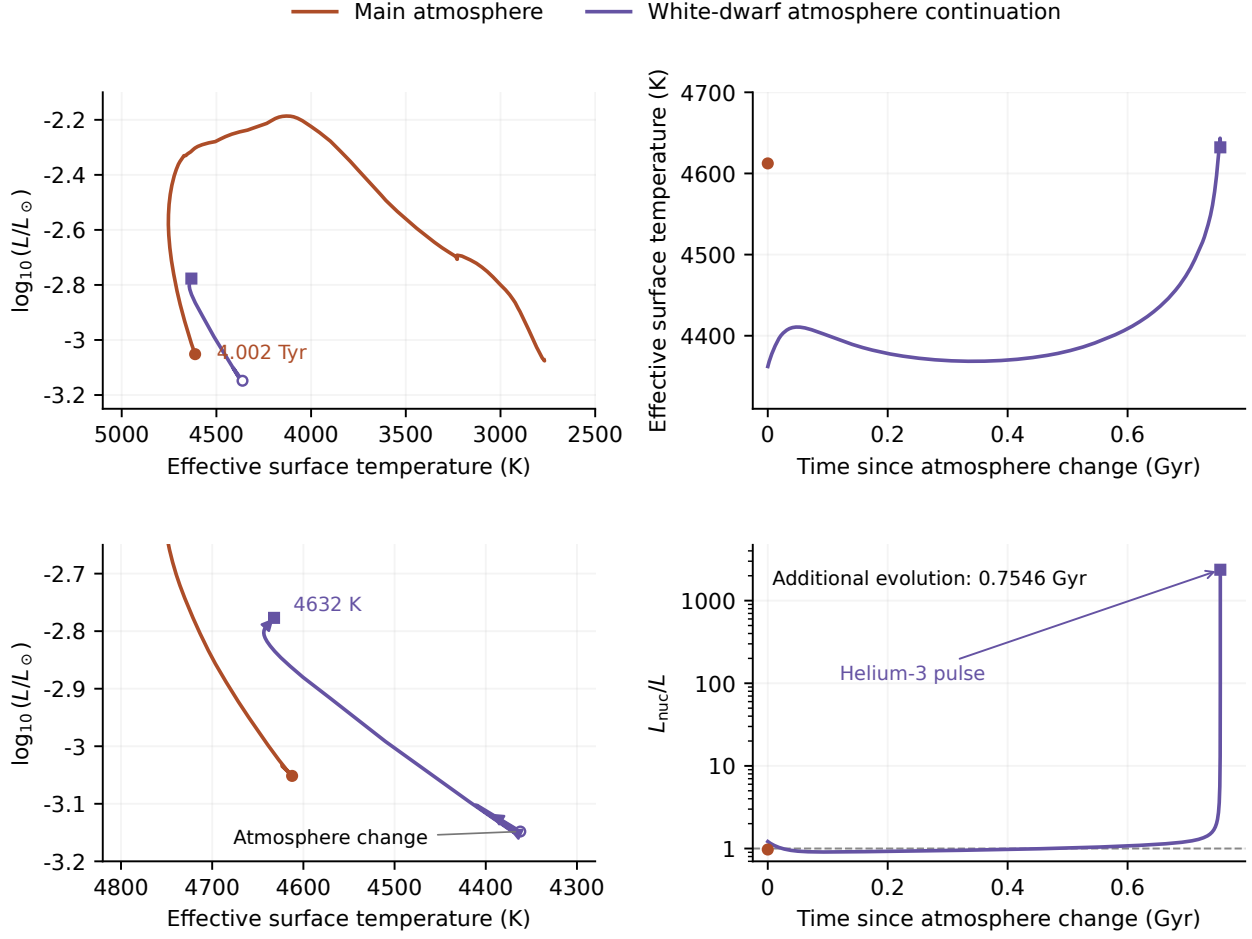


Figure 5: Track S with two atmosphere prescriptions. Orange: the TLUSTY families through 4.002Tyr. Purple: the continuation with the pure-hydrogen white-dwarf table, which differs from the TLUSTY boundary by 2.6% in temperature and 49% in pressure at the joining depth in the sampled 4600 K, $\log g = 5.9$ atmosphere, so the initial offset is a change of boundary condition. Left: the HR track and an enlarged view of the adjustment. Right: surface temperature and the ratio of nuclear to photon luminosity against time since the switch (logarithmic scale below). The continuation covers 0.7546 Gyr in 252 accepted intervals and ends as a helium-3 shell becomes convective.

5.2 What is established

The pulse forms on every mass grid tried (578, 666, 848, 1695, 1763, 1899, 1923, 1983, 2010 and 2115 points, all refined around the burning layer) and with both instantaneous and finite convective mixing. Nuclear power rises from a quiescent $3 \times 10^{30} \text{ erg s}^{-1}$ to $1\text{--}2 \times 10^{36} \text{ erg s}^{-1}$ while the photon luminosity stays near $7 \times 10^{30} \text{ erg s}^{-1}$: the released energy goes into the structure. Hydrostatic evolution remains valid; at the coarse-grid endpoint the local heating time $c_P T / \epsilon_{\text{nuc}}$ is 472.4 yr against a dynamical time of 23.46 s. The shell is only partly degenerate. Before shell convection the most productive cell, at $m/M = 0.9612$ and 10.72 MK, has thermal expansion coefficient $\delta = 0.9099$, close to the ideal-gas value, a logarithmic temperature sensitivity of the heating rate of 16.52, and a thickness 0.05541 times

its radius; thin-shell geometry and strong temperature sensitivity support a thermal instability without strong degeneracy. The same mechanism has been found in slowly accreting white dwarfs, where helium-3 from incomplete pp burning ignites a convective layer before a nova outburst [19]; the fuel supply and structure differ, so the precedent supports the mechanism, not the amplitude.

Through 3422 yr on the 666-point grid the accepted nuclear powers integrate to 7.964×10^{45} erg, about 5% of the 1.561×10^{47} erg that burning the whole helium-3 inventory would release; surface radiation carries about 7×10^{41} erg over the same interval. Nuclear energy plus escaping neutrinos agrees with the change in nuclear rest mass. During the pulse the first-law residual is measured against the larger of nuclear and photon luminosity, with a relative limit of 2×10^{-8} .

5.3 What is not established

Spatial convergence. Table 7 and Figure 6 show that the amplitude and timing of the burning are not converged. At equal remaining helium-3, the finite-mixing calculations on 578 and 666 points differ in nuclear power by a factor of 2.901; over a matched 100 yr on 1927 and 1983 points the deposited energy differs by 8.261% and the final power by 15.24%; subdividing the burning layer fourfold (1695 to 1923 points) over 688.5 yr reduces the deposited energy by 55.17%. Later, during the decline, successive refinements agree to 0.5–1.5%, which supports the shell-burning phase but not the onset or the peak. Inserting mass points causes a mechanical readjustment with an initial energy residual of 1.844×10^{43} erg, identified as geometric rather than physical. Convective regions that grow across the composition gradient merge abruptly (near 3006 and 3341 yr on the finest grids), and substep tests across the mergers agree to 0.01–0.3% in integrated energy while a gradual-mixing variant differs by 10.17%; the internal timing of each merger is unresolved.

Table 7: Flash calculations from the common post-switch model. Time is measured from the common profile before shell convection; “peak” means the largest value reached, not a converged maximum.

Points	Mixing	Interval	Peak L_{nuc} (erg s ^{−1})	Max T (MK)
578	instantaneous / finite	1647 yr	—	15.30
666	instantaneous / finite	2139 yr	2.671×10^{35}	15.29
666	finite, energy flux on cell faces	3429 yr	5.537×10^{36}	21.86
848	finite, energy flux on cell faces	3342 yr	4.835×10^{36}	21.73
1695	finite, energy flux on cell faces	6803 yr	3.281×10^{35}	15.39
1983 (F1)	finite from onset, all convective layers	5.744 Myr	—	—
2115 (F2)	instantaneous then finite	42.48 Myr	—	—

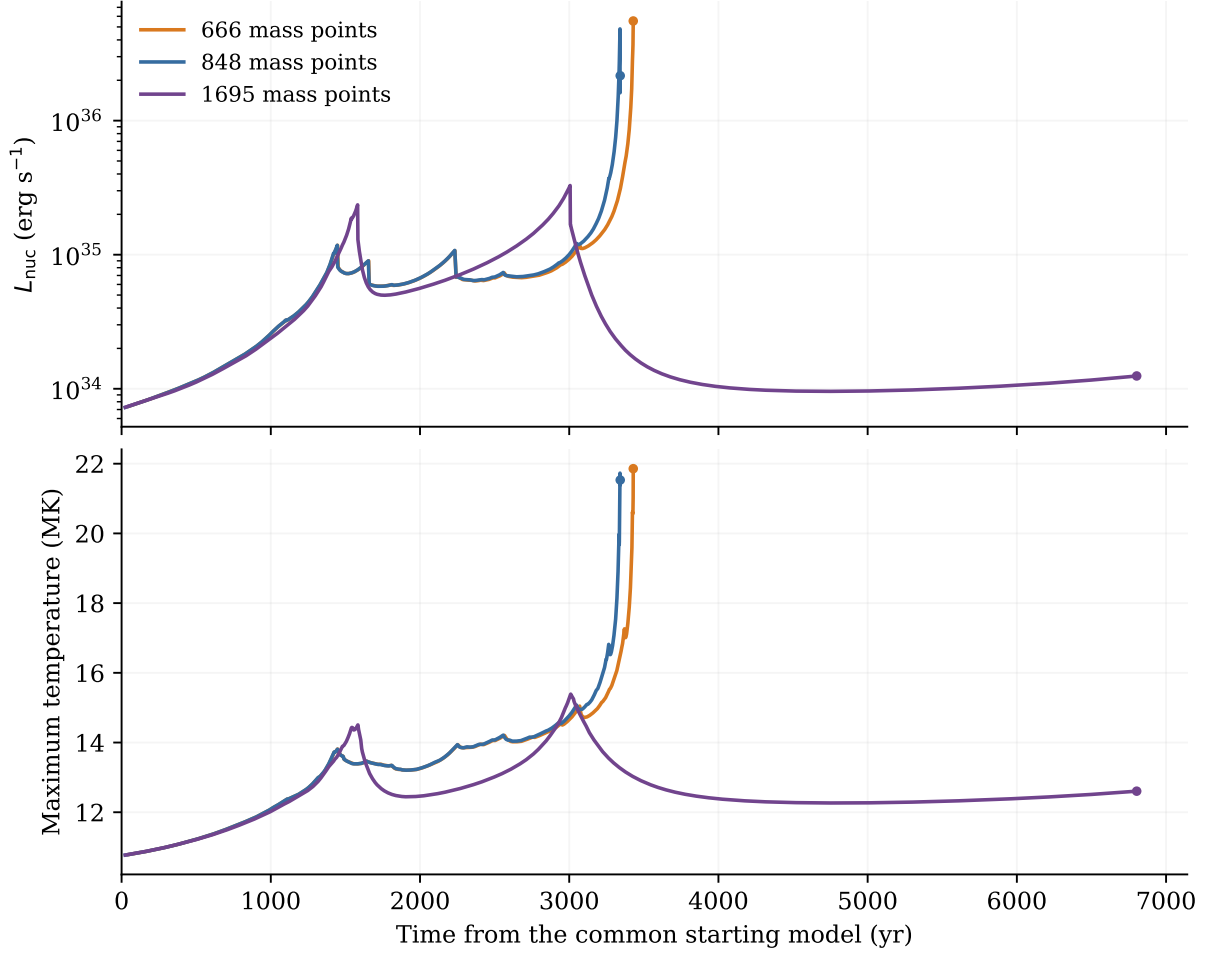


Figure 6: Nuclear luminosity and maximum internal temperature on three mass grids, with total energy and composition enthalpy transported across the same mass surfaces. All use the white-dwarf table boundary and finite convective mixing without semiconvection; time runs from the common profile before shell convection. The largest sampled shell temperatures are 21.86, 21.73 and 15.39 MK; the finest grid reaches 12.60 MK after a long decline and modest reheating. Sharp decreases near 3006 and 3341 yr are convective-region mergers. Their largest sampled nuclear powers differ by more than an order of magnitude; the peak is not converged.

Dependence on the boundary prescription. Both flash continuations inherit the white-dwarf table’s normalization, which differs from the TLUSTY boundary by 2.6–3.0% in temperature and 49–91% in pressure at the states sampled. Applying the calculated mixed-to-hydrogen composition ratios to the hydrogen table retains that normalization rather than testing it. Comparisons of two extrapolations of the table below $\log g = 5.5$, over 10^3 – 2×10^6 yr, change the endpoint temperature by at most 11.27 K and the deposited nuclear energy by at most 0.03%; these test the extension, not the absolute boundary. The pulse develops 0.7546 Gyr after the switch, but that delay does not show that it is independent of the switch: the shell was fading before it, and the LBA97 model, the grey MESA sequences and Ember’s own grey-boundary sequence (Section 6.2) show no flash. Whether ignition survives the original atmosphere can be decided only by a comparison that starts before the histories diverge. A pure-hydrogen TLUSTY source rectangle at optical depth 100 above $\log g = 5.95$ is being computed for that control; the 4800 K, $\log g = 6.0$ columns are already validated, and the control is pending.

Mixing physics. The flash calculations set semiconvective and thermohaline coefficients to zero. The

reaction that powers the flash lowers the mean molecular weight and can drive thermohaline mixing, as in red giants [21]; whether it does so strongly enough to alter the shell requires the composition gradient and transport times to be assessed. Holding the mixing coefficient at its beginning-of-step value during each implicit solve, as MESA does [22], passes the species and energy checks with successive changes of 0.09470%, 0.04722% and 0.02357% in integrated nuclear energy, but pointwise composition differences do not decrease.

5.4 The two continuations

Two histories were carried through the pulse from the same post-switch model (Figure 7, Table 8). F1 applies finite composition transport throughout all convective layers from the onset of shell convection. F2 first mixes the cooler convective layers instantaneously, then uses finite transport once the shell convection connects to the surface envelope; its accumulated thermal and composition history therefore differs from F1’s, and local checks of its later evolution do not establish that it is F1’s outcome. In F2 the envelope becomes helium-enriched when the convective regions connect: over a checked 0.6656 yr the surface hydrogen falls from 0.9984 to 0.8085 while helium-3 rises to 0.007264, yet 99% of the metals remain inside $m/M = 0.4456$, so the atmosphere can be helium-rich while remaining nearly metal-free. Both histories pass a radius maximum before their temperature maximum: expansion is followed by contraction during which the surface heats while the luminosity falls, consistent with delayed structural adjustment. At their endpoints both are declining in temperature but remain 33–56% nuclear-powered, with most of the remaining helium-3 outside the burning layer. In F2 at 1.345×10^4 yr the outer convective envelope, then 4.033% of the mass, holds 86.44% of the remaining helium-3 but supplies 0.7504% of the nuclear power; between 1.443×10^4 and 1.701×10^4 yr it loses only 0.01453% of its helium-3 while the active layers deplete. The fuel that could power a later episode is therefore separated from the burning region, and its consumption depends on transport that these calculations do not yet resolve; fuel depletion alone does not exclude another unstable episode, and neither history is a white-dwarf cooling sequence.

Table 8: Endpoints of the two flash continuations. Times from the common model before shell convection; both provisional.

	F1 (1983 points)	F2 (2115 points)
Time computed	5.744 Myr	42.48 Myr
Saved T_{eff} maximum	6422 K at 1.139 Myr	6399 K at 1.088 Myr
Endpoint T_{eff}	5903 K	4771 K
Endpoint nuclear power (erg s^{-1})	1.126×10^{31}	3.166×10^{30}
Nuclear share of surface luminosity	32.84%	55.85%
Endpoint radius	$0.09048 R_{\odot}$	$0.05633 R_{\odot}$
Surface X , X_3	—	0.8200, 0.007167
Remaining hydrogen	—	$0.003792 M_{\odot}$
Central T , ρ	—	6.192 MK, $3.932 \times 10^4 \text{ g cm}^{-3}$

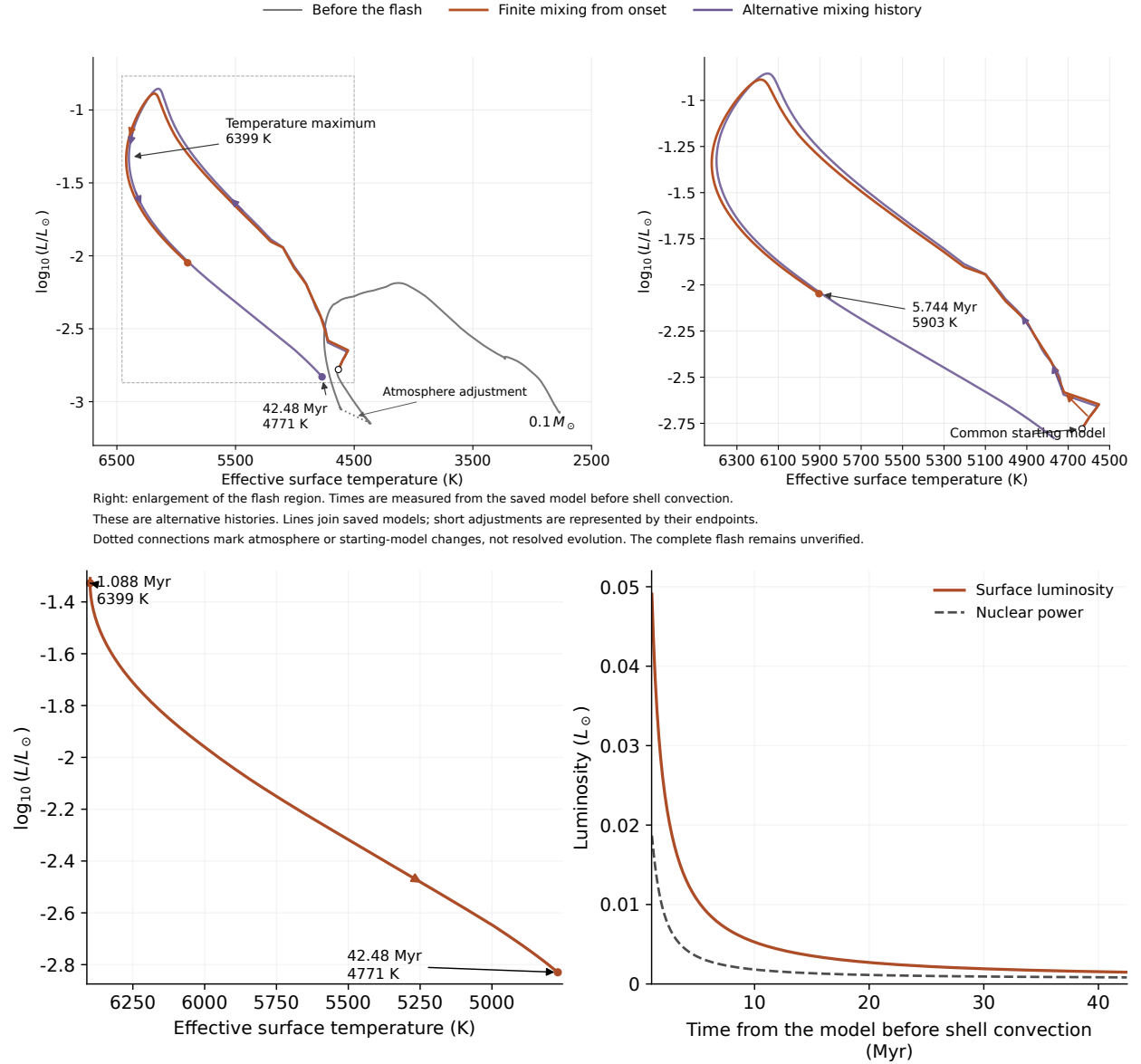


Figure 7: The two flash continuations. Upper: HR tracks of F1 (orange, finite mixing from onset) and F2 (purple, alternative mixing history), with the preceding Track S evolution in gray; dotted connections mark the atmosphere switch and starting-model adjustments, not resolved evolution. The right panel enlarges the flash region. Lower: F2 on 2115 points, an enlarged HR diagram around its temperature maximum (left) and photon luminosity with deposited nuclear power against time (right). Both histories use the white-dwarf table boundary switched in at 4.002Tyr; whether the flash occurs with the original atmosphere has not been tested. Agreement of the two HR tracks does not establish convergence of the peak power or of the internal mixing.

At 29.48 Myr in F2 the reduced pp network produces helium-3 at $1.263 \times 10^{12} \text{ g s}^{-1}$ and destroys it at $4.106 \times 10^{11} \text{ g s}^{-1}$, and helium-3 fusion supplies 23.12% of the nuclear power, down from 26.97% five million years earlier; summing the saved sources reproduces the inventory change to better than 10^{-10} . This establishes the production–destruction balance, not stability against a later flash. In F2 at 12.07 Myr, 46.16% of the nuclear power comes from ${}^3\text{He}+{}^3\text{He}$, 49.94% from the hydrogen reactions that build ${}^3\text{He}$ and 3.901% from the reduced ppII branch; the helium-3 inventory is 82.97% of its value in

the common starting model. Between 2.012×10^5 and 2.512×10^5 yr, nuclear burning supplies 31.97% of the integrated surface emission and structural adjustment 68.03%. The atmosphere tables entered by the continuations (hydrogen–helium mixtures at 4500–6500 K and $\log g = 4.2$ –5.8) were checked for depth and interpolation to better than 0.2% in temperature and 1.1% in pressure (Appendix A); the absolute hydrogen normalization remains the unresolved uncertainty.

6 Comparisons

6.1 The published LBA97 model

LBA97 [9] followed the full hydrogen-burning evolution of a $0.1M_\odot$ star into white-dwarf cooling with a grey atmosphere, approximate opacity changes as hydrogen is consumed, and the interior opacities and reaction rates of the time. Table 9 compares their stated values with Track P, which is the closest in physics. The starting states are defined differently: LBA97 includes contraction and defines arrival on the main sequence by nuclear reactions supplying the whole luminosity, after about 2 Gyr, whereas Track P starts from a static hydrogen-burning model at age zero. That offset is small compared with the later differences.

Table 9: LBA97 (literature values; curve readings marked $\simeq$) and Track P.

Quantity	LBA97	Track P
Initial surface temperature (K)	2228	2766
Initial luminosity ($L_\odot$)	$\simeq 0.000417$	0.0008362
Initial central density (g cm^{-3})	309	381.7
Maximum ^3He mass fraction	0.0995	0.1003
Age at maximum ^3He (Tyr)	1.38	0.7103
Radiative layer first forms (Tyr; central X)	5.742 ($\simeq 0.16$)	3.546 (0.1751)
Age at hydrogen fraction 0.2034 (Tyr)	$\simeq 5.6$	3.40
Highest surface temperature (K)	5807	≥ 5546 (endpoint, still rising)

Ember initially radiates 2.006 times the published luminosity, and the ages at maximum ^3He differ by the ratio 1.943. That is what the nuclear clock requires: with $0.07M_\odot$ of hydrogen and about 6×10^{18} erg g^{-1} deposited, the energy supply is of order 8×10^{50} erg and the burning time is that supply divided by the mean nuclear luminosity, so a persistently lower luminosity lengthens the clock. A homology estimate gives the same scale. With $\rho_c \propto R^{-3}$, $T_c \propto R^{-1}$, $\epsilon_{pp} \propto \rho T^\nu$ for $\nu \simeq 4$ –6, and nuclear power equated to $4\pi R^2 \sigma T_{\text{eff}}^4$,

$$R \propto T_{\text{eff}}^{-4/(\nu+5)}, \quad L \propto T_{\text{eff}}^{4(\nu+3)/(\nu+5)} \simeq T_{\text{eff}}^{3.1-3.3},$$

and the temperature ratio predicts a luminosity ratio of 1.959–2.029. The thermal times $GM^2/(RL)$ are only 3.0 and 5.5 Gyr, so a starting thermal imbalance cannot sustain a trillion-year offset; the difference lies in the equilibrium sequence that the physics maintains. The ratios change later in the evolution, and no single rescaling of time aligns the two calculations.

The low LBA97 luminosity was recognized in 1997: their Section 3.1 reports a poorer fit to the observed mass–luminosity relation than the Burrows models. The independent solar-metallicity model of Chabrier and Baraffe [10] at $0.1M_\odot$ and 1 Gyr gives 2811 K, $0.1240R_\odot$ and $0.0008511L_\odot$, close to Ember; they also find that a grey Eddington boundary generally makes low-mass models hotter, so the grey atmosphere alone does not explain LBA97’s cooler surface. Identifying the cause needs controlled substitutions of atmosphere, opacity and equation of state, which have not been done. A reconstruction of the

original Fortran program is available for such tests but does not reproduce LBA97: with two low-temperature opacity choices it flags a nonconvective centre at 3.307 and 2.946Tyr (against the published 5.742), and switching only that opacity changes its luminosity by 54.5%. Its one close agreement, a highest surface temperature of 5936 K against 5807 K, does not validate the rest. Recovering Track P’s photospheric density and temperature at optical depth 2/3 and plotting them over an opacity map, as in LBA97’s Figure 6, shows the expected drift to hotter, thinner photospheres and nothing that changes the comparison.

The ^3He maximum agrees to about 1%, but Ember reaches it in half the time and consumes hydrogen faster throughout (Figure 3). The HR track alone conceals this because it does not show how long the star spends at each position. Candidate causes are the frequency-dependent atmospheres, composition-specific opacities, the equation of state and the updated pp rates; one-ingredient-at-a-time comparisons are needed to separate them. Agreement with LBA97 is not a measure of accuracy.

6.2 Independent calculations with MESA

We evolved the same mass and bulk composition with the unmodified MESA release 26.4.1 [1] in two configurations. Neither shares Ember’s equation of state, opacities, network or atmosphere, so they test whether an independent code produces the same sequence of events, not the accuracy of any one ingredient.

Grey-atmosphere configuration. MESA starts from pre-main-sequence contraction with its supplied pp/CNO network, OPAL and low-temperature opacities, electron conduction, an Eddington grey atmosphere and mixing length 2. Nuclear burning reaches 99% of the surface luminosity at 951.3 Myr (3076 K). From that model the evolution was continued with and without microscopic diffusion. Without diffusion, central hydrogen crosses 0.001 at 2.372Tyr, the surface reaches its maximum temperature of 6766 K at 2.406Tyr, and the model cools to 595.9 K by 2.564Tyr, where the atmosphere requests temperatures below the 501.2 K floor of the Ferguson opacity table. The final state has $R = 0.02572R_\odot$, $L = 7.516 \times 10^{-8}L_\odot$, surface $X = 0.08778$ and $0.0009172M_\odot$ of hydrogen. This is a computed cooling branch under a grey boundary; it says nothing about that boundary’s accuracy for a cold helium-rich atmosphere. With diffusion, the same configuration reaches only 1.225Tyr (3471 K, central $X = 0.4593$) in 29.91 CPU-hours: the atmospheric equation-of-state inversion fails repeatedly on unphysical trial pressures, a numerical fault rather than a gap along the accepted structure.

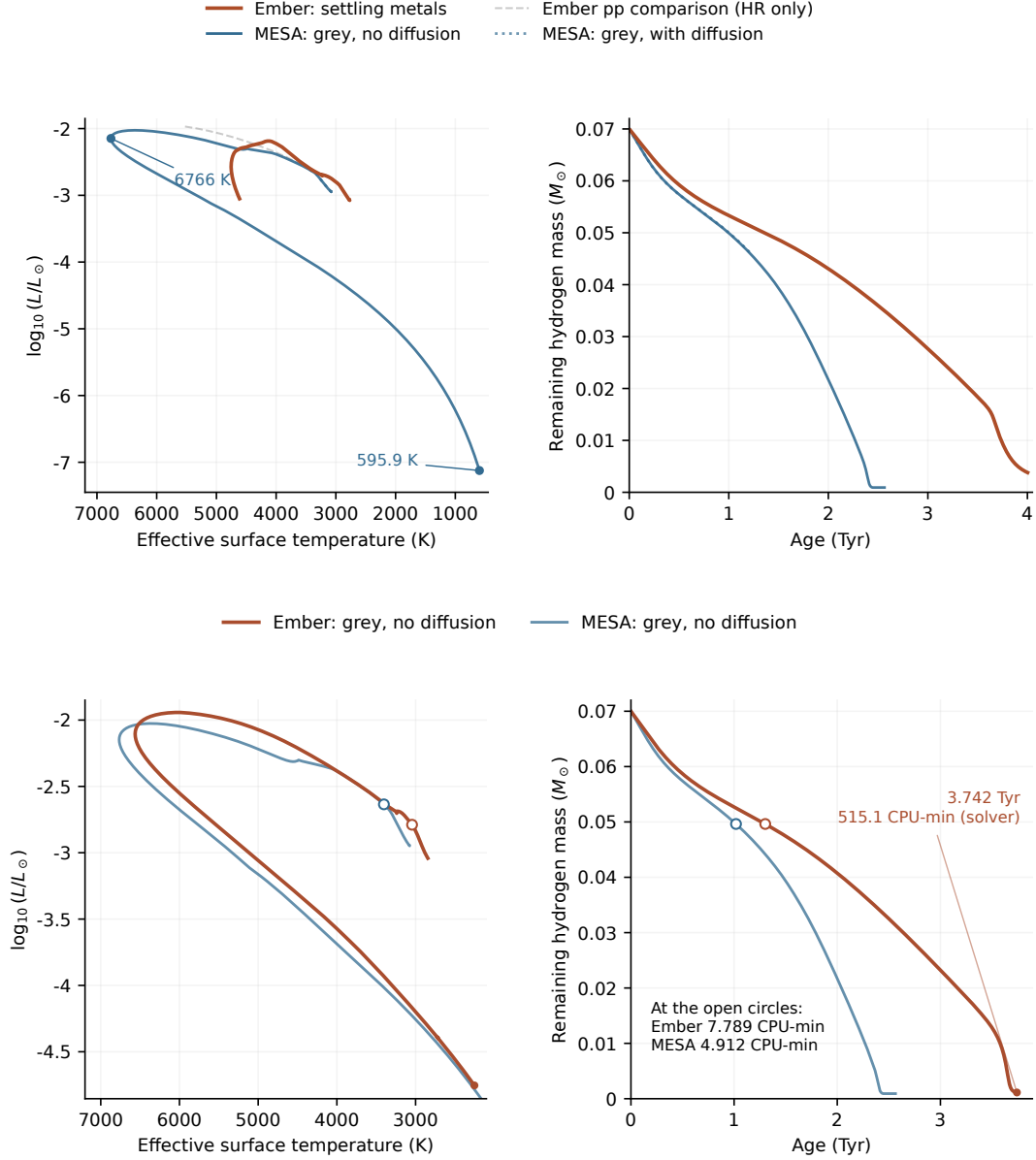


Figure 8: Ember and MESA at $0.1 M_{\odot}$. Upper left: HR tracks, including the computed grey MESA cooling branch; upper right: remaining hydrogen mass. Solid blue is MESA without diffusion, dotted blue the same grey configuration with diffusion through 1.225Tyr; orange is Track S and faint dashed gray Track P. Lower: the grey-boundary comparison without diffusion, Ember (orange, through 3.742Tyr) against MESA (blue); open circles mark equal remaining hydrogen, $0.04963 M_{\odot}$, and the filled point the last Ember model. MESA ages include contraction; Ember ages start at the static hydrogen-burning model. Equations of state, opacities and networks differ throughout.

Ember with the same grey boundary. For a like-for-like comparison, an Ember run uses the same Eddington grey boundary and no microscopic diffusion, with convection, pp+CN burning and Ember’s own equation of state and opacities (Figure 8). It reaches 3.742Tyr at 2252 K and central $X = 7.991 \times 10^{-7}$, with a conductive core at 1.971 MK under a convective envelope. Both codes show the same transition when a radiative layer separates the core from the envelope: a luminosity dip and a

decline in central helium-3 once replenishment from the envelope stops. Ember passes it at 3241 K and 3.305Tyr with central $X = 0.1728$; MESA at 4498 K and 2.295Tyr with $X = 0.08286$. The difference in transition temperature and composition is unexplained; the material physics and time resolution differ, and the transients have not been compared at matched accuracy. Below 3471 K the Ember grey run includes the quantum correction to nuclear screening of Potekhin and Chabrier [34, 35], which changes a matched 100 Myr by 0.08920 K. Neither this sequence nor either MESA sequence develops a helium-3 flash.

At equal remaining hydrogen ($0.04963M_{\odot}$) Ember is at 1.302Tyr, 3045 K and $0.001626L_{\odot}$, and MESA at 1.019Tyr, 3403 K and $0.002316L_{\odot}$. Their helium-3 inventories differ (0.005740 and $0.003710M_{\odot}$), so equal hydrogen does not mean equal energy released. To that point Ember used 7.789 solver CPU-minutes and MESA 4.912 process CPU-minutes, a ratio of 1.586; the counters have different scope, and a fully convective phase does not measure relative cost during shell burning or cooling. Ember’s grey sequence used 515.1 solver CPU-minutes through 3.742Tyr and MESA’s grey cooling sequence 1.727 CPU-hours through 750.6 K; these are different phases of different calculations and their ratio is not a benchmark.

Tabulated-atmosphere configuration. The fuller MESA configuration uses PHOENIX and Castelli–Kurucz atmosphere tables, OPLIB and Ferguson opacities, the blend of FreeEOS, OPAL–SCVH, HELM and Skye equations of state, conduction with the Blouin hydrogen/helium correction, 34 species with individual diffusion, Ledoux convection with mixing length 1.9 and Brown–Garaud–Stellmach thermohaline mixing. It has reached only 362 Myr of contraction (2986 K). Diagnostic runs trace the slowness to diffusion acting on the thin radiative skin above the convective envelope: when the diffusion domain becomes active it homogenizes the skin and moves the atmospheric boundary abruptly. Deepening the diffusion boundary to 10^5 K and adding a small mixing coefficient in the outer 10^{-6} of the mass lets a diagnostic run continue to 2.986Tyr (4694 K, central $X = 7.123 \times 10^{-6}$, surface $X = 0.9903$) without retries. These runs identify a numerical cause; they do not supply a physical surface-mixing prescription and are not plotted.

A further diagnostic switched a hydrogen-rich MESA model at 3.163Tyr (4737 K, $\log g = 6.010$) from the grey boundary to the supplied white-dwarf table. Within 1000 yr the photon luminosity fell by 19.96% and the effective temperature by 5.410%; over the following 33.65 Myr the model reached 4176 K with a 15.33% larger radius. Below $\log g = 6$ the MESA wrapper blends the table with a grey boundary (22.81% grey weight at the endpoint), so this is not a pure table continuation. It is the same kind of boundary change that precedes Ember’s flash, and it shows that the response is an evolving thermal adjustment rather than an equilibrium offset.

What the comparison establishes. An independent code with different physics produces the same sequence: full convection, a radiative layer with a luminosity dip, central hydrogen exhaustion, a temperature maximum and a turn toward cooling. The ages differ by about a third between the codes, in the same direction as the LBA97 difference but smaller; given how strongly the nuclear clock depends on luminosity this is expected rather than diagnostic. The 50 Gyr diffusion comparison of Section 4 shows that element redistribution, not the omitted oxygen branches, is the stronger candidate for the late separation between the Ember tracks, although a unique division has not been measured. Finally, the complete pp/CNO network in MESA does not by itself make the calculation expensive.

7 What the model still lacks, and what it would change

The following omissions are the ones that can change an interpretation in this paper. For each, the measured size of the effect is given where one exists.

Atmospheres below 4600 K, above $\log g = 6.1$, and without hydrogen. Track S stops where its TLUSTY families stop, and the flash continuations rely on a hydrogen table whose normalization differs from TLUSTY by 49% in joining pressure (Section 5). Cooling through 1000, 500 and 100 K requires dense, cool atmospheres with nonideal chemistry, pressure-dependent and collision-induced absorption, and refraction [16]. A hydrogen-free atmosphere is not yet possible: the source programs normalize abundances to hydrogen, and a test at 4017 K shows that rescaling all number abundances by one factor preserves composition but can change individual absorption coefficients by a factor of 27.3. These gaps decide whether a computed cooling age exists at all; nothing in Sections 4–5 is a cooling age.

Grains. Equilibrium chemistry finds no condensates in the 3000–3200 K atmospheres tested nor in the hydrogen-poor structures through 6400 K, but finds them in the cool upper layers of 2600–2800 K models; dusty models become relevant near 2800 K and below [14]. Removing the condensed species from the gas at 2800 K changes the joining pressure by 0.1246% and temperature by -0.02347% . A grain-opacity calculation on one fixed 2800 K profile, with a condensed mass fraction of 1.094×10^{-4} in $0.1 \mu\text{m}$ compact grains, gives integrated optical depths near $1 \mu\text{m}$ of 0.002610 (absorption) and 0.002829 (scattering); coupling grains to the atmospheric structure and thermodynamics has not been done. Grains are a possible correction during contraction, the cool initial main sequence and the eventual cold-remnant phase. The continuous model reaches the main sequence near 2763 K, so the grain-free assumption also needs assessment there.

Carbon conversion and molecular opacity. A fixed metal mixture misrepresents the cool opacity once carbon has been converted to nitrogen, because CO formation couples the available carbon and oxygen [33]. Complete conversion of the GS98 carbon captures a hydrogen mass fraction of 5.728×10^{-4} . At $X = 0.20$ and $\rho = 5.995 \times 10^{-6} \text{ g cm}^{-3}$ the Rosseland absorption mean changes by the ratios in Table 10; across densities of 10^{-8} – $10^{-4} \text{ g cm}^{-3}$, half conversion raises the 2705 K mean by 18.67–30.22%. In a coupled 3200 K atmosphere at $\log g = 5.15$, half conversion raises the joining temperature by 1.601% and lowers the pressure by 5.681%; applied to the 3.400Tyr structure for 10 Myr this changes the luminosity by -5.162% and the effective temperature by -1.361% . At 5400 K the response is negligible. The effect is material for the cool part of the track and must be included in a track that carries converted catalysts to the surface.

Table 10: Rosseland absorption mean relative to the fixed GS98 mixture at $X = 0.20$ and $\rho = 5.995 \times 10^{-6} \text{ g cm}^{-3}$ (gas only).

T (K)	Half the carbon converted	All the carbon converted
2705	1.225	1.407
3521	1.151	1.263
4583	0.9971	0.9912
5964	0.9901	0.9744

Microscopic transport in degenerate and partially ionized matter. At the centre of the 3.890Tyr Track P model the temperature is 0.1506 of the electron Fermi temperature, so degeneracy enters the transport coefficients, and the thermal-diffusion and heat-flow coefficients must use one consistent definition [17, 23]. A coupled ion/electron calculation on the saved gradients, including electron–electron collisions and the leading ion recoil term, gives provisional outward hydrogen velocities of $(1.101\text{--}1.303) \times 10^{-9} \text{ cm s}^{-1}$ at enclosed mass 0.6972, with local drift times $H_P/|w_H|$ of 12.78–15.13 Gyr there and 3.694–4.244 Gyr near mass fraction 0.9738; the thermal force contributes 23.63–24.48% of the drift. The range spans two screening choices and is not an uncertainty. Including electron–electron collisions gives 0.5277–0.8780 of the elastic-only conductivity across the hot nonconvective region. These are the coefficients Tracks S and F use above about 2 MK; below that the tracks represent species exchange by convection alone, and cold radiative layers still lack a supported transport law. Heat carried by species motion is at most 0.09557% of the local luminosity on hot nonconvective faces; its omission below 2 MK changes surface luminosity by at most 0.01320% over 5 Myr.

Interior opacity sources. The hot interior opacity comes from one archive, and independent archives disagree at the few-percent level. On 1940 states with $X = 0$ or 0.05 and $Z = 0.01$ or 0.03, the Los Alamos OPLIB means distributed with MESA [37] differ from the TOPS means used here by a median of 2.364% and by up to 7.279%; on 15 warm-envelope layers the Opacity Project [36] differs from OPLIB by -9.730 to $+4.619\%$, and from TOPS by 2.165% at a sampled state. The archives use different element sets and atomic physics, and no track has been run with the alternatives. Where the sampled layers are convective or conductive the stellar response to opacity changes of this size is unresolved (Table 6); in the radiative envelope of the depleted star a $\pm 10\%$ change moves the effective temperature by 3.216 K, so a few-percent archive difference is below the other uncertainties but is not zero.

The cold helium interior. The energy equation used throughout is $L_\gamma = L_{\text{nuc,dep}} + L_{\text{grav}} - L_{\nu,\text{thermal}}$ with nuclear neutrinos removed from $L_{\text{nuc,dep}}$; at 3.890Tyr in Track P the terms are 0.01077, 3.201×10^{-5} and (nuclear neutrinos) $3.362 \times 10^{-4} L_\odot$, with plasma losses of $1.260 \times 10^{-7} L_\odot$ and a discrete residual below 8×10^{-11} . Beyond the computed tracks the remnant needs a consistent liquid and solid helium free energy with quantum and mixture effects [15]: specifying melting at $\Gamma = 175$ does not fix the latent heat or the liquid heat capacity. The cold-electron thermal response, electron and ionic entropies and liquid/solid free energies have been implemented and checked against thermodynamic identities and against Ioffe EOSFI22 (largest density-response discrepancy 1.05×10^{-6} in ideal-ion pressure units), but they are not yet combined into the production equation of state. Crystallization and its latent heat, the depth at which convection couples to the isothermal core, and pycnuclear rates in the actual helium-rich composition [32] all remain to be evaluated.

Omitted reactions, neutrinos and mixing. Pep, hep, ppIII and the oxygen branches are absent in every track; adding a closed CN cycle to a Track P state adds 2.381% to the deposited pp energy and lowers the surface luminosity by 0.1239% after 5 Myr. Only plasma neutrino losses are included. Semiconvective heat flux, thermohaline mixing during the flash, rotation, magnetism, mass loss and accretion are absent. The tracks describe an isolated, nonrotating star; accretion in particular would change the surface composition on which the atmosphere depends [20].

External heating and encounters. Encounters, accretion and particle decay are treated in Section 8 as conditional extensions of the isolated track [11]. For close encounters, energy transferred into stellar oscillations is distinct from retained heat; gravitational radiation can remove mode energy [12], and the thermalized fraction must be determined before it enters a heating history.

8 Lifetime timeline

The timeline runs from formation to the conditional disappearance of the remnant. Computed entries are labelled by track and clock; the rest are literature values or conditional stages whose ages remain to be calculated. Several late processes can overlap, and disappearance means loss of the baryonic remnant through decay products.

Age or condition	Event	Basis
Before $t = 0$	Formation and accretion	Cloud collapse, disk and outflows set the mass, entropy and angular momentum; outside the fixed-mass calculation. Low-mass pre-main-sequence models provide the comparison [26].
Contraction clock, 0–1 Gyr	Deuterium burning and contraction	Computed: 2763 K and $0.1265R_{\odot}$ near 1 Gyr with 99.09% nuclear support; continued to 3.562Tyr (Section 3). Lithium depletion is not followed.
Long-term clock, $t = 0$	Static hydrogen-burning model	Tracks S, F and P start here with $X = 0.7$, $T_{\text{eff}} = 2766$ K.
0.7103Tyr (P)	Maximum ${}^3\text{He}$	Central mass fraction 0.1003, then destruction overtakes production. LBA97: 0.0995 at 1.38Tyr.
3.546–3.551Tyr (P, F, S)	Loss of full convection	A radiative layer separates centre and envelope at central $X \simeq 0.175$; the centre becomes radiative near 3.551Tyr in F and S. LBA97: 5.742Tyr, $X \simeq 0.16$.
3.650–3.767Tyr (S)	Central hydrogen depletion	Central X crosses 10^{-3} , 10^{-4} and 10^{-5} ; $0.0147M_{\odot}$ of hydrogen remains at the first crossing and burns outside the centre.
3.673, 3.838Tyr (S)	Luminosity and temperature maxima	$0.006513L_{\odot}$; 4754 K. Burning still dominates; no cooling sequence is established.
4.002Tyr (S)	Last computed model	4612 K, central $X = 3.528 \times 10^{-8}$, 97.28% nuclear.
Provisional	Helium-3 shell flash	Occurs after a boundary switch; dependence on the original atmosphere untested (Section 5).
To be calculated	Residual burning fades	Record nuclear heat below 10%, 1% and 0.1% of the photon luminosity, including any recovery. LBA97 reaches 5807 K at its turn; a comparison value only.
To be calculated	Diffusion and separation	Hydrogen collects at the surface while heavier species settle; a helium core does not imply a helium atmosphere.
To be calculated	Molecules, grains, dense atmospheres	Follow molecular and collision-induced absorption, condensation and settling where local conditions permit [14, 16].
To be calculated	Crystallization and separation	Helium and mixture phase equilibria, latent heat, solid heat capacity; the front follows density and core temperature, not surface temperature [15].
To be calculated	Envelope and core couple	Whether and when the convective envelope reaches the nearly isothermal interior; order relative to crystallization unknown.
$T_{\text{eff}} = 1000, 500, 100$ K	Cold-remnant milestones	Record age, core temperature, radius, composition, solid fraction and power sources; below 100 K continue with a solid, partly neutral description.
$\gtrsim 10^{14}$ yr	Changing environment	Retained and ejected objects in evolving stellar and dark-matter environments; accretion of gas, debris and solid bodies is conditional; no unique encounter rate is assumed.
Close passages	Tidal heating, capture, collision	Distinguish heat from spin and oscillations; couple an encounter survey to waiting-time distributions and post-pulse cooling.

Age or condition	Event	Basis
Allowed interactions	Dark-matter heating	Compare capture, thermalization, evaporation and annihilation with the photon luminosity; include a no-annihilation case [31].
Nuclear times competitive	Density-driven fusion	Pycuclear and impurity reactions against cooling and decay times in the helium-rich composition [32].

Decay and disappearance (conditional). Proton decay has not been detected; the Super-Kamiokande partial-lifetime limit for $p \rightarrow e^+ \pi^0$ is $\tau_p/B > 2.4 \times 10^{34}$ yr [27], and baryogenesis does not require it [28]. Virtual black holes supply a possible channel with $\tau_p \sim 10^{45}$ yr [30], distinct from the far slower collective tunnelling of degenerate matter [29]. We therefore carry finite-lifetime scenarios and a stable-baryon branch rather than one disappearance date. In order of condition, the stages are: decay heating exceeds cooling and environmental power, possibly long before much mass is lost; accumulated decays change the composition and mass; degeneracy is lost and the mass-radius relation must be recalculated for the actual composition; decay products and then thermal radiation escape as columns shrink, so that an optically thin object need not radiate as a blackbody; the remnant ceases to be bound, which is distinct from the decay of every former baryon; and finally discrete survival replaces a continuous mass. In the stable-baryon branch cooling, nuclear and environmental alternatives continue without a decay-imposed endpoint.

A reference clock for the decay era. If every baryon has the same constant, independent mean lifetime τ_B , with no accretion or stable daughters, and t_d measures time from the start of the comparison,

$$\langle M_B(t_d) \rangle = M_{B,0} e^{-t_d/\tau_B}, \quad L_{\text{heat}} = f_{\text{dep}}(t_d) \frac{\langle M_B \rangle c^2}{\tau_B}, \quad (3)$$

where the deposited fraction f_{dep} may evolve and a physical channel would also account for daughter masses and binding. Half the mass is lost at $t_d = 0.6931 \tau_B$, 90% at $2.303 \tau_B$, 99% at $4.605 \tau_B$ and 99.99% at $9.210 \tau_B$. For N_0 independently decaying baryons,

$$\Pr(t_{\text{last}} \leq t_d) = \left(1 - e^{-t_d/\tau_B}\right)^{N_0}, \quad \langle t_{\text{last}} \rangle = \tau_B H_{N_0}, \quad (4)$$

with H_N the harmonic number. For $0.1 M_\odot$, $N_0 \simeq 1.2 \times 10^{56}$, the mean last decay occurs near $129.7 \tau_B$ with a central 90% interval of 128.0 – $132.1 \tau_B$. This dates the last of the original baryons, dispersed or not, not the survival of a bound star. These are analytic consequences of the stated assumptions, not an extrapolation of any computed track; a reference lifetime is not a measured age, and changing it changes temperature, chemistry and energy escape as well as the time scale.

9 Computing cost and reproducibility

The model sequences in Figure 2 used 47.49 CPU-hours on an Apple M4 Max: 36.31 for Track S, 10.98 for Track F and 2.259 for Track P, with 861 shared states (2.052 CPU-hours) counted once. These totals include table loading, step-doubling comparisons and adaptive retries within the runs that supply the plotted curves, and exclude independent controls and discarded tracks. Track S occupied 15.13 hours of elapsed time excluding pauses between continuations. Building the atmosphere, opacity and equation-of-state tables that the tracks interpolate cost about 2.6 times more, roughly 122 CPU-hours for the documented jobs; a rough conversion at 2×10^9 operations per CPU-second, uncertain by a factor of

three, puts the sequences near 3×10^{14} floating-point operations and the tables near 10^{15} . The continuous Hayashi-started sequence through 3.562Tyr uses 8.788 CPU-hours on four threads, including its retained contraction and main-sequence intervals and process startup; separate atmosphere and accuracy controls are excluded. The flash continuations were computed in segments; a complete cooling-track runtime has not been measured, and none of these numbers predicts it.

Implicit transport is required: the explicit stability limit of the initial diffusion coefficients is 672.7–936.1 yr, while accepted steps late on Track S have a median length of 59.44 Myr among those longer than 1 Myr. Reusing exactly matching nuclear evaluations reduced the CPU time of a matched run through 3.757Tyr from 162.4 to 101.8 minutes with identical output; the other performance changes recorded during development were each measured in isolation and are not additive.

The repository at <https://github.com/oklo/ember> contains the code, table-generation instructions, the records of input choices and checks, this paper and the numerical data behind its figures. The LBA97 comparison records the selected image coordinates, axis calibration and the checksum of the original PDF, so its curves can be rebuilt without the image. Full raw outputs, generated physical tables, saved stellar states and executable archives remain local; regenerating the bulk tables is required before a stellar calculation can be run from a fresh checkout. Instructions and input manifests for recreating the figures accompany the paper in the repository.

A Technical reference

Opacity and equation-of-state continuations

Settling raises the core metal fraction above the $Z = 0.03$ limit of the tabulated ATOMIC opacities. Through $Z = 0.08$ the radiative opacity is continued linearly from the $Z = 0.02$ and 0.03 sources at fixed temperature, density and hydrogen abundance,

$$\kappa(T, \rho, X, Z) = \kappa(T, \rho, X, 0.03) + \frac{Z - 0.03}{0.01} [\kappa(T, \rho, X, 0.03) - \kappa(T, \rho, X, 0.02)], \quad (5)$$

with all source temperature and density bounds enforced. Transferring the OPLIB composition ratio from $Z = 0.02$ to a withheld $Z = 0.03$ source differs from the ATOMIC value by at most 0.1867% at 34 core states, and the linear and ratio forms differ by 0.001212% in luminosity over a matched 5 Gyr. Independent OPLIB data through $Z = 0.2$ support linear scaling within 3.302% at 6–10 MK; in the region with $Z > 0.06$ radiation carries at most 0.04154% of the gradient-driven heat. As the envelope falls below $Z = 0.01$ the molecular opacity interpolates the actual AESOPUS values, including $Z = 0$ and 0.004; at higher temperatures κ is continued linearly in Z from the 0.01 and 0.02 sources with a floor at $Z = 10^{-4}$. Hydrogen dependence is continued in $\log \kappa$ from the $X = 0.70$ and 0.75 sources through pure hydrogen; the molecular source ends at $1 - X = 10^{-7}$. In the dense hydrogen envelope, where the OPLIB table ends at $\log R = 1.5$ ($R = \rho/T_6^3$), the TOPS hydrogen slope is blended in over $\log R = 1.25$ – 1.45 and used to $\log R \leq 1.8$ for $\log T = 5.6$ – 6.1 ; convection carries 99.96% of the heat at the affected layer. In dense core cells without spectral coverage (0.125–0.45 keV, 10^4 – 10^5 g cm $^{-3}$, $X \leq 0.2$) the tabulated uncut opacity is multiplied by an inferred plasma correction; radiation carries at most 0.2535% of the local transport there.

The variable-metal equation-of-state family covers $Z = 0$ – 0.08 (0.16 for the core) and all 23 sampled Track S structures. Against direct FreeEOS evaluations the maximum differences are 0.03028% in pressure, 0.09056% in adiabatic gradient, 0.7642% in heat capacity and 1.037% in Γ_1 ; 36 low-density and 28 high-metal comparisons agree within 0.009197% and 0.02530% in the material responses. The Track F family (152 tables, 38 hydrogen and four ^3He fractions) has a maximum second-composition-

derivative difference of 0.1499% at 102 compositions and a heat-capacity difference of 0.1062% on the saved 512-point structure. Composition derivatives are evaluated without an abundance floor, including when a species is exactly absent; 128 analytic comparisons and five FreeEOS curvature checks at zero ^3He (within 0.007181%) pass. The continuous driver’s potential matches direct FreeEOS to 0.02589% in pressure and 0.02467% in internal energy on the 512-zone reference profile.

Atmosphere families

Table 12 lists the atmosphere tables entered by Track S and the flash continuations. “Depth” is the largest change in joining temperature and pressure when the source column is computed to a deeper lower boundary; “interpolation” is the largest difference between an independently solved column and the interpolated value. Missing cells remain inaccessible to the evolution rather than being filled.

Table 12: Atmosphere tables along Track S and the flash continuations.

Composition	T_{eff} (K), $\log g$	Columns	Depth (T ; P)	Interpolation (T ; P)
Track P reference: $X = 0.1\text{--}0.7$, $X_3 = 0\text{--}0.12$, $Z = 0.02$	to 5600; 4.9–5.4	220	0.007883%	0.7399%; 0.5707%
Warm extension, $X = 0.15, 0.2$	5400–6400; 5.7, 6.0	300	0.009372%	0.4860%
Late Track F grid, $X = 0.70\text{--}0.90$, $X_3 = 0\text{--}0.03$	3400–4000; 4.9–5.15	48	—	1.837% in T
Metal response $Z = 0.01\text{--}0.02$ (paired)	to 4400; 5.025	pairs	0.0002576%	0.6093%; 3.064%
$Z = 0.005\text{--}0.01$	to 4400; 5.1	pairs	—	1.671%; 1.980%
$Z = 0.0025\text{--}0.005$	to 4400; 5.1	12 pairs	0.001301%; 0.002243%	0.03091%; 0.2311%
$Z = 0.001\text{--}0.0025$	4000–4400; 5.1	8 pairs	—	0.2539%; 3.111%
$Z = 0.0005\text{--}0.001$	4000–4400; 5.1	8 pairs	—	2.031%; 2.755%
$X = 0.9\text{--}0.98$, $Z = 0.001$, $X_3 = 0$, 0.003	4000–4800; 4.9–5.4	70	—	0.004095%; 0.01689%
$X = 0.995\text{--}0.9995$, $Z = 10^{-6}\text{--}10^{-4}$	4600–5200; 5.15–5.65	94	—	0.09719%; 4.656%
$1 - X = 10^{-5}\text{--}10^{-3}$, $Z = 10^{-8}\text{--}10^{-6}$	4600–5200; 5.4–5.65	81	(shallowest ends at $\tau = 168.3$)	0.4624%; 0.3930%
$Z \leq 10^{-6}$, He 10^{-10}	4700–6000; 5.5–5.95	72	4.327 $\times$ $10^{-6}\%$; 4.770 $\times$ $10^{-5}\%$	0.1751%; 0.5986%
Zero metal, cold	4650–4800; 5.9–5.95	6	below 10^{-6}	0.1154%; 1.108%
Gravity continuation	4650–4800; 5.95–6.1	inferred	—	0.1878%; 2.391% (9 solved)
Temperature continuation	4600; 5.9	1 solved	—	within 0.05%
H–He mixtures (flash), $X = 0.78\text{--}0.9955$	4500–5500; 4.95–5.80	54	0.02452%; 0.03703%	0.08784%; 1.094%
H–He mixtures, three X	4700–6000; 4.95–5.25	36	0.01078%; 0.06911%	—
H–He mixtures, three X	6000–6500; 4.20–5.55	60	0.01373%; 0.09599%	—

Composition	T_{eff} (K), $\log g$	Columns	Depth (T ; P)	Interpolation (T ; P)
H–He mixtures, three X	5000–6000; 5.4–5.8	27	—	$7.587 \times 10^{-6}\%$; 0.1582%
MESA hydrogen white-dwarf table [2]	2000–40,000; 5.5–9.5	table	joined at $\tau = 25.12$	extrapolated below $\log g = 5.5$

Within $0.90 < X < 0.94$ the logarithms of the joining values are interpolated in hydrogen between the $X = 0.90$ and 0.94 families; the metal response is calibrated at $X_3 = 0$ and applied separately from the small isotope response, and helium below a mass fraction 0.005 is neglected in the 4600–4800 K, $\log g = 5.9$ –6.1 domain (solved helium-bearing columns change the joining values by less than 0.1% in T and 2.5% in P). In the flash continuations the hydrogen table is extrapolated below $\log g = 5.5$ using either its lowest two rows or the gravity response of solved near-hydrogen atmospheres at $\log g = 4.35$ and 4.50; the two inferred references differ by up to 2.189% in temperature and 15.04% in pressure at tabulated points, and paired stellar comparisons between them are those quoted in Section 5.

Time-step and conservation checks

Every accepted interval compares one full step with two half steps in structure, species, total hydrogen and energy. Species balances are checked independently in each composition region together with the inert-metal mass, nuclear rest-mass release and the first law. For Tracks S, F and P, the relative time-error budgets are 0.0003 for radius, density and temperature, 0.003 for local ^3He and 0.001 for the other species; the abundance solver tolerance is 10^{-10} with a stricter fallback (a 10^{-12} control changes mass fractions by at most 1.360×10^{-10}). During the flash, deposited nuclear energy is integrated consistently with the implicit equations, time refinement permits 0.5% in deposited energy and 0.1% in structure and endpoint nuclear power (a 1% allowance was tested and rejected because a local temperature difference of $|\Delta \ln T| = 0.001091$ exceeded the declared 0.001 limit), and the first-law residual is measured against the larger of nuclear and photon luminosity with a limit of 2×10^{-8} . Mesh insertions preserve species inventories, internal energy and the nodal gravitational-energy sum. Increasing the separate abundance-change limit per step from 0.001 to 0.005 saves 24.99% of CPU over a matched 1 Gyr with a luminosity change of 0.0005727%; a larger structure tolerance was rejected (Section 2). Electron-pair collision integrals for the late shell-burning calculation cover degeneracy parameter $\eta = -6$ –64 and screening parameter $b = 0.1$ –2400, with interpolation and integration checks within 0.3466%. Collision integrals use 96 Jacobi and 128 Legendre points; against 384/512 points the maximum scaled coefficient difference is 4.851×10^{-9} .

References

- [1] Jermyn, A. S., et al. (2023). Modules for Experiments in Stellar Astrophysics (MESA): Time-Dependent Convection, Energy Conservation, Automatic Differentiation, and Infrastructure. *ApJS*, 265, 15. <https://arxiv.org/abs/2208.03651>.
- [2] MESA documentation. Table atmospheres. <https://docs.mesastar.org/en/latest/atm/table.html>.
- [3] Rohrmann, R. D., Althaus, L. G., & Kepler, S. O. (2011). Lyman α wing absorption in cool white dwarf stars. *MNRAS*, 411, 781–791. doi:10.1111/j.1365-2966.2010.17716.x.
- [4] Colgan, J., et al. (2016). A New Generation of Los Alamos Opacity Tables. *ApJ*, 817, 116. doi:10.3847/0004-637X/817/2/116.

- [5] Los Alamos National Laboratory. TOPS opacity service, ATOMIC/New elemental library and mixture queries. aphysics2.lanl.gov/opacity/lanl/.
- [6] Davis, Y. T., et al. (2024). The EBLM Project XII. An eccentric, long-period eclipsing binary with a companion near the hydrogen-burning limit. [arXiv:2312.09156v2](https://arxiv.org/abs/2312.09156v2), Table 2 and abstract.
- [7] Agol, E., et al. (2021). Refining the Transit-timing and Photometric Analysis of TRAPPIST-1: Masses, Radii, Densities, Dynamics, and Ephemerides. [arXiv:2010.01074](https://arxiv.org/abs/2010.01074), Table 7.
- [8] Ribas, I., et al. (2017). The full spectral radiative properties of Proxima Centauri. [arXiv:1704.08449](https://arxiv.org/abs/1704.08449), Table 11.
- [9] Laughlin, G., Bodenheimer, P., & Adams, F. C. (1997). The End of the Main Sequence. *ApJ*, 482, 420. [doi:10.1086/304125](https://doi.org/10.1086/304125).
- [10] Chabrier, G., & Baraffe, I. (1997). Structure and evolution of low-mass stars. *A&A*, 327, 1039. [arXiv:astro-ph/9704118](https://arxiv.org/abs/astro-ph/9704118), Table 2.
- [11] Adams, F. C., & Laughlin, G. (1997). A dying universe: the long-term fate and evolution of astrophysical objects. *Rev. Mod. Phys.*, 69, 337. [doi:10.1103/RevModPhys.69.337](https://doi.org/10.1103/RevModPhys.69.337).
- [12] Rathore, Y., Blandford, R. D., & Broderick, A. E. (2005). Resonant excitation of white dwarf oscillations in compact object binaries—I. The no back reaction approximation. *MNRAS*, 357, 834. [Published calculation](#).
- [13] Haft, M., Raffelt, G., & Weiss, A. (1994). Standard and nonstandard plasma neutrino emission revisited. [arXiv:astro-ph/9309014](https://arxiv.org/abs/astro-ph/9309014).
- [14] Chabrier, G., Baraffe, I., Allard, F., & Hauschildt, P. (2000). Evolutionary models for very low-mass stars and brown dwarfs with dusty atmospheres. *ApJ*, 542, 464. [arXiv:astro-ph/0005557](https://arxiv.org/abs/astro-ph/0005557).
- [15] Baiko, D. A., & Chugunov, A. I. (2022). *Ab initio* thermodynamics of one-component plasma for astrophysics of white dwarfs and neutron stars. [arXiv:2112.04822](https://arxiv.org/abs/2112.04822).
- [16] Blouin, S., Dufour, P., & Allard, N. F. (2018). A new generation of cool white dwarf atmosphere models. I. Theoretical framework and applications to DZ stars. [arXiv:1807.06616](https://arxiv.org/abs/1807.06616).
- [17] Beznogov, M. V., Potekhin, A. Y., & Yakovlev, D. G. (2016). Diffusive heat blanketing envelopes of neutron stars. *MNRAS*, 459, 1569. [doi:10.1093/mnras/stw751](https://doi.org/10.1093/mnras/stw751).
- [18] Thoul, A. A., Bahcall, J. N., & Loeb, A. (1994). Element diffusion in the solar interior. *ApJ*, 421, 828. [arXiv:astro-ph/9304005](https://arxiv.org/abs/astro-ph/9304005).
- [19] Denissenkov, P. A., Herwig, F., Bildsten, L., & Paxton, B. (2013). MESA Models of Classical Nova Outbursts: The Multicycle Evolution and Effects of Convective Boundary Mixing. *ApJ*, 762, 8. [Published article](#).
- [20] Bauer, E. B., & Bildsten, L. (2019). Polluted White Dwarfs: Mixing Regions and Diffusion Timescales. *ApJ*, 872, 96. [arXiv:1812.09602](https://arxiv.org/abs/1812.09602).
- [21] Charbonnel, C., & Zahn, J.-P. (2007). Thermohaline mixing: A physical mechanism governing the photospheric composition of low-mass giants. *A&A*, 467, L15. [arXiv:astro-ph/0703302](https://arxiv.org/abs/astro-ph/0703302).
- [22] Paxton, B., et al. (2011). Modules for Experiments in Stellar Astrophysics (MESA). *ApJS*, 192, 3. [Published methods](#).
- [23] Paxton, B., et al. (2018). Modules for Experiments in Stellar Astrophysics (MESA): Convective Boundaries, Element Diffusion, and Massive Star Explosions. [arXiv:1710.08424](https://arxiv.org/abs/1710.08424), section 3.2.2.
- [24] Adelberger, E. G., et al. (2011). Solar fusion cross sections. II. The pp chain and CNO cycles. *Reviews of Modern Physics*, 83, 195.
- [25] Acharya, B., et al. (2025). Solar fusion III: New data and theory for hydrogen-burning stars. *Rev. Mod. Phys.*, 97, 035002. [doi:10.1103/8lm7-gs18](https://doi.org/10.1103/8lm7-gs18).

- [26] Baraffe, I., Homeier, D., Allard, F., & Chabrier, G. (2015). New evolutionary models for pre-main sequence and main sequence low-mass stars down to the hydrogen-burning limit. [arXiv:1503.04107](#).
- [27] Takenaka, A., et al. (Super-Kamiokande Collaboration, 2020). Search for proton decay via $p \rightarrow e^+\pi^0$ and $p \rightarrow \mu^+\pi^0$ with an enlarged fiducial volume in Super-Kamiokande I–IV. [arXiv:2010.16098](#).
- [28] Fukugita, M., & Yanagida, T. (1986). Baryogenesis without grand unification. *Phys. Lett. B*, 174, 45. [doi:10.1016/0370-2693\(86\)91126-3](#).
- [29] Adams, F. C., Laughlin, G., Mbonye, M., & Perry, M. J. (1998). The gravitational demise of cold degenerate stars. *Phys. Rev. D*, 58, 083003. [arXiv:astro-ph/9808250](#).
- [30] Adams, F. C., Kane, G. L., Mbonye, M., & Perry, M. J. (2001). Proton decay, black holes, and large extra dimensions. [arXiv:hep-ph/0009154](#).
- [31] Bell, N. F., et al. (2024). Heavy Dark Matter in White Dwarfs: Multiple-Scattering Capture and Thermalization. [arXiv:2404.16272](#).
- [32] Gasques, L. R., et al. (2005). Nuclear fusion in dense matter: Reaction rate and carbon burning. *Phys. Rev. C*, 72, 025806. [arXiv:astro-ph/0506386](#).
- [33] Lederer, M. T., & Aringer, B. (2009). Low temperature Rosseland opacities with varied abundances of carbon and nitrogen. [Published article](#); [arXiv:0810.5672](#).
- [34] Potekhin, A. Y., & Chabrier, G. (2012). Thermonuclear fusion in dense stars: Electron screening, conductive cooling, and magnetic field effects. *A&A*, 538, A115. [arXiv:1201.2133](#).
- [35] Chugunov, A. I., & DeWitt, H. E. (2009). Nuclear fusion reaction rates for strongly coupled ionic mixtures. *Physical Review C*, 80, 014611. [arXiv:0905.3844](#).
- [36] Seaton, M. J. (2005). OP data on CD for mean opacities and radiative accelerations. [doi:10.1111/j.1365-2966.2005.00019.x](#).
- [37] Farag, E., et al. (2024). An Expanded Set of Los Alamos OPLIB Tables in MESA: Type-1 Rosseland-mean Opacities and Solar Models. *The Astrophysical Journal*. [doi:10.3847/1538-4357/ad4355](#).